

Universal distance modes from DESI BAO and Type Ia supernovae: what do cosmological rulers actually measure?

MATIAS ZALDARRIAGA¹

¹*Institute for Advanced Study, Princeton, NJ 08540, USA*

ABSTRACT

We use an SVD decomposition of the low-redshift distance measurements from DESI BAO and three Type Ia supernova compilations to identify the leading linear directions probed by the data and to localize the tension with the Λ CDM CMB-anchored predictions. The leading direction V_0 — whose data amplitude we denote c_0 — is, to high accuracy, a measurement of $\Omega_m h^2$: the projection of the data on V_0 probes essentially this one CMB-derived parameter combination. BAO constrains this parameter more tightly than the CMB itself; the three SN compilations do not. In every extension of Λ CDM we consider, the leading measurable direction remains V_0 , and it is where most of the tension with the CMB resides. In the w_0 – w_a extension a second direction V_1 becomes measurable and provides an independent test of dynamical dark energy; the data show no significant tension in this direction. The only other beyond- Λ CDM extension that opens a genuinely new measurable direction is spatial curvature, and only marginally and only for BAO; both measurable directions then independently prefer positive spatial curvature, though the second direction is poorly constrained.

Keywords: cosmology: observations — dark energy — methods: statistical — large-scale structure of Universe

1. INTRODUCTION

The Dark Energy Spectroscopic Instrument (DESI) has measured baryon acoustic oscillations (BAO) across $0 < z < 2.3$ with unprecedented precision (DESI Collaboration 2025a). Combined with cosmic microwave background (CMB) data and Type Ia supernovae (SNe Ia), the DESI collaboration reports a 2.8 – 3.9σ preference (depending on SN dataset) for evolving dark energy in the w_0 – w_a parameterization (DESI Collaboration 2025b). What drives this preference? We show that it arises primarily from what could be interpreted as a $\sim 2\sigma$ discrepancy in $\Omega_m h^2$ between the CMB and low-redshift distance data, rather than from any dark energy signal.

There are many ways to search for deviations from Λ CDM. Here we take Λ CDM as a well-established baseline whose parameters are constrained by the CMB (we use ACT DR6; ACT Collaboration et al. 2025) to a small allowed range. The question we ask is: *how large is the space of deviations that low-redshift distance probes can actually detect?*

The answer is simpler than one might expect. The CMB-allowed Λ CDM parameter range maps into a low-dimensional subspace of distance predictions. For both

BAO and SN measurements, the effective dimensionality of this map is one: a single SVD direction V_0 along which the CMB allows essentially all of the model motion. We write c_0 for the projection of the data onto V_0 . The matrix being decomposed has one row per CMB chain sample and one column per whitened observable (Section 2.3); its leading singular value exceeds the next by a factor of ~ 25 for BAO. The result depends only on the CMB parameter uncertainties and the low-redshift measurement covariance — not on the low-redshift data values.

We can then ask the same question for extensions beyond Λ CDM: does a given extension (dynamical dark energy, spatial curvature, etc.) open a genuinely new measurable direction in observable space, beyond V_0 ? Again, this is answerable from the theory and covariance alone.

The result is a set of predetermined directions onto which we project the data. Searching for deviations reduces to reading off a few numbers (tensions in units of σ , one per mode).

Preview of results. All tension concentrates in a single universal direction V_0 : BAO shows $+2.2\sigma$ in c_0 against the CMB prediction, while the three SN compilations are individually mild (DES-Dovekie -0.8σ , Pan-

theon+ -1.1σ , Union3 -1.7σ) and do not have enough constraining power in c_0 to confirm or refute the BAO signal. The independent dark-energy direction V_1 , which becomes measurable in the w_0 - w_a extension, shows only $+1.2\sigma$ for BAO — consistent with a cosmological constant. The only other extension that opens a measurable new direction is spatial curvature, and only for BAO.

Among the other extensions we considered, the phenomenological lensing rescaling A_{lens} — a multiplicative factor on the smoothing of CMB acoustic peaks by gravitational lensing, with $A_{\text{lens}} = 1$ in Λ CDM (Calabrese et al. 2008) — is notable: it reduces the tension by shifting the predicted c_0 mean and increasing its variance, reminding us of the fact that the CMB constraint on $\Omega_m h^2$ has a lensing component. The physics is the same as the well-known effect of shifting the reionization optical depth τ : in the CMB, $A_s e^{-2\tau}$ controls the lensing amplitude, so a high- τ universe and a high- A_{lens} universe are nearly degenerate (Sailer et al. 2026). In their analysis Sailer et al. (2026) make this explicit: dropping Planck’s low- ℓ polarization and letting the rest of the CMB+BAO data float τ within Λ CDM yields $\tau = 0.090 \pm 0.012$, ~ 3 – 5σ above the Planck low- ℓ value. Since essentially all current cosmological analyses simply adopt the Planck τ measurement, a systematic in that single number would propagate identically into essentially every CMB-anchored cosmological analysis.

Relation to previous work. Several recent analyses have examined the DESI results from angles closely related to ours. Efstathiou (2025a) rotates the BAO distance measurements into directions parallel and perpendicular to the Planck Λ CDM degeneracy. The parallel direction effectively measures ω_m , showing a $\sim 2.1\sigma$ tension with Planck—in quantitative agreement with our $+2.2\sigma$ BAO c_0 tension. The perpendicular direction and the equation of state at the pivot redshift, $w(z_{\text{piv}} = 0.5) = -0.996 \pm 0.046$, show no departure from Λ CDM, consistent with our finding that the dark energy direction c_1 is null ($+1.2\sigma$). Efstathiou (2025b) has also identified potential photometric systematics in the DES Y5 supernova compilation, finding a ~ 0.04 mag offset between low and high redshift subsamples; we use the recalibrated DES-Dovekie sample (Popovic et al. 2025) which partially addresses these concerns. Our DES-Dovekie c_0 tension is indeed the mildest of the three SN datasets (-0.8σ) but none is significant.

Ye & Lin (2025) independently confirm the $\sim 2\sigma$ BAO–CMB tension using Bayesian suspiciousness, and find that the phantom crossing redshift $z_{\text{pc}} = 0.45 \pm 0.04$ coincides with the best-constrained scale—matching our BAO c_1 pivot $z = 0.46$. The pivot redshift approach was pioneered by Cortès & Liddle (2024) in the con-

text of DESI DR1, who identified what they term the “PhantomX coincidence”: that w crosses -1 precisely at the epoch of observation. Our SVD analysis provides an explanation for this coincidence: the apparent $w \neq -1$ at the c_0 pivot is $\Omega_m h^2$ tension dressed in dark energy clothing, while the genuinely independent dark energy direction (c_1) is consistent with a cosmological constant at its own pivot. Note that c_0 , although where the tension lives, is a near-perfect proxy for $\Omega_m h^2$ — a parameter that the CMB also constrains tightly along the same combination — so the tension can in principle be resolved by any mechanism that shifts the CMB inference of $\Omega_m h^2$ (e.g., τ , A_{lens} ; Section 7). The dark-energy direction c_1 , by construction, cannot be moved by such mechanisms — which makes its null tension at $+1.2\sigma$ a genuine constraint on dynamical dark energy.

Our V_0 direction is the empirical, multi-probe counterpart of the analytically constructed Model-independent Expansion Discrepancy Index (MEDI) of Weiner (2026): both single out the same parameter combination from very different starting points. The MEDI identifies $\omega_m r_d^2$ as the combination that acoustic-scale measurements constrain; within Λ CDM, where r_d is nearly fixed by early-universe physics, this reduces to $\Omega_m h^2$, which is why c_0 is almost purely an $\Omega_m h^2$ mode. The physical origin of the $\Omega_m h^2$ dominance in BAO was recognized by Eisenstein & White (2004) and Knox (2006). Appendix A provides our own derivation connecting the log derivatives of distance observables to the β -vector coefficients.

The curvature analysis in Section 6 connects to Chen & Zaldarriaga (2025), who find $\Omega_k = +0.0023 \pm 0.0011$ from standard MCMC methods. We did not run a dedicated Ω_k MCMC chain in this work; we use simple formulas to estimate the value of Ω_k that the BAO data prefer.

Finally, Sailer et al. (2026) argue that a systematic in Planck’s low- ℓ polarization τ would act as a single-point failure for essentially all current CMB analyses, and that the rest of the CMB+BAO data within Λ CDM prefer $\tau = 0.090 \pm 0.012$, a ~ 3 – 5σ shift from the low- ℓ measurement. Their τ channel and our A_{lens} channel involve the same physics expressed differently: $A_s e^{-2\tau}$ controls the CMB lensing amplitude, so A_{lens} and τ are partially degenerate.

The paper is organized as follows. Section 2 describes the data and SVD methodology. Section 3 establishes the universality of c_0 across all probes. Section 4 presents the c_0 tension measurements. Section 5 reinterprets the w_0 - w_a preference. Section 6 examines spatial curvature. Section 7 considers other CMB-side extensions. Section 8 discusses the implications and summa-

rizers our conclusions. Appendix A provides an analytical derivation of the c_0 universality.

This paper, both calculations and draft, was written using Claude Code. It is based on some old calculations done at the time that the original DESI paper appeared and partially presented in a talk at Perimeter Institute¹. The goal of those calculations was just for me to better understand the claims and are presented here in case anyone finds them interesting. Creating this manuscript served as a guide for me to understand how to use the new AI tools. I make no claims of originality or novelty. All code, data-loading scripts, and the full analysis pipeline needed to reproduce every number and figure in this paper are available in a public GitHub repository.² It contains the SVD analysis package, the scripts that turn the public DESI DR2 BAO, supernova, and CMB MCMC chains into the distance-ratio decompositions used here, the marimo notebook that regenerates every figure, and a machine-readable registry of all claims and quoted numbers; the aim is that a reader can re-derive the results from the public inputs, inspect each numerical choice, and freely modify or extend the analysis. This manuscript will not be submitted to any journal. It has been sufficiently “refereed” by Claude Code³. I am happy to correct any mistakes and of course anyone is free to download the code and ask any agent to fix or improve it. Hopefully this disclaimer is sufficient to appease the arxiv police⁴, but maybe transparency will trigger a ban and you will have to wait a year for any new paper with my name on it.

2. DATA AND METHOD

2.1. Data

We use three classes of low-redshift distance data, summarized in Table 1: DESI DR2 BAO measurements, three SN compilations (Union3, Pantheon+, DES-Dovekie), and the ACT DR6 / Planck 2018 CMB chains we use as the high-redshift reference. Each is described in turn below.

2.1.1. DESI DR2 BAO

We use the official DESI DR2 BAO measurements from DESI Collaboration (2025a). The data consist of 13 observables spanning $z = 0.295$ to $z = 2.330$: one isotropic measurement (D_V/r_d) from the Bright Galaxy

Survey (BGS) at $z = 0.295$, plus six anisotropic redshift bins each contributing two observables (D_M/r_d and D_H/r_d), for a total of $1 + 6 \times 2 = 13$ observables from Luminous Red Galaxies (LRG), Emission Line Galaxies (ELG), Quasars (QSO), and Lyman- α tracers.

The BAO observables are ratios of distances to the sound horizon. The comoving transverse distance is $D_M(z) = c \int_0^z dz'/H(z')$ (for the spatially flat models considered here; Ω_k is reinstated in Section 6), the Hubble distance is $D_H(z) = c/H(z)$, and the volume-averaged distance is $D_V(z) = [z D_M(z)^2 D_H(z)]^{1/3}$. For the supernovae the observable is the distance modulus $\mu(z) = 5 \log_{10}[d_L(z)/10 \text{ pc}]$, with luminosity distance $d_L(z) = (1+z) D_M(z)$. Here r_d is the comoving size of the sound horizon at the baryon drag epoch ($z_d \approx 1060$), the scale imprinted on the baryon distribution that calibrates the BAO feature (DESI Collaboration 2025a; ACT Collaboration 2025). It differs from the sound horizon at last scattering, r_* (sometimes written $r_{s,*}$), which sets the CMB acoustic scale θ_* ; the two have different numerical values, and throughout we use the drag-epoch quantity r_d to normalize the BAO distances. Our fiducial value is $r_d = 147.09 \text{ Mpc}$.

The full 13×13 covariance matrix is block-diagonal in redshift — measurements in different redshift bins are uncorrelated, while within each anisotropic bin D_M/r_d and D_H/r_d are jointly fit to the same 2D BAO feature and so are strongly anti-correlated ($r \approx -0.4$).

It is worth quantifying how tightly the CMB pins these distance ratios at high redshift. The acoustic scale $\theta_* = r_*/D_M(z_*)$, evaluated at the last-scattering redshift z_* , is constrained to 0.025% across the ACT ΛCDM chain used below, so $D_M(z_*)/r_*$ is essentially fixed. The BAO observables differ only in normalizing by the drag-epoch horizon r_d rather than r_* and—for D_V —in carrying an extra factor of the Hubble rate. Propagating these through the chain, $D_M(z_*)/r_d$ has only 0.032% scatter (the drag and last-scattering horizons are tightly correlated) and $D_V(z_*)/r_d$ only 0.061%, both far below DESI’s percent-level measurement errors. This is the quantitative reason the distance-ratio curves must return to the fiducial cosmology at high redshift, whatever the low-redshift expansion history.

2.1.2. Type Ia Supernovae

We analyze three SN compilations: Union3 (Rubin et al. 2023) with 22 pre-binned redshift bins ($z = 0.05$ – 2.26), Pantheon+ (Scolnic et al. 2022) with 1701 individual SNe ($z = 0.001$ – 2.26), and DES-Dovekie (Popovic et al. 2025) with 1755 SNe ($z = 0.025$ – 1.14). All cos-

¹ <https://pirsa.org/25080017>

² <https://github.com/matiaszaldarriaga/desi-w0wa-svd>

³ This is a joke, but I am definitely of the camp that wants to let LLMs “cook” (<https://arxiv.org/abs/2602.10181>)

⁴ <https://xcancel.com/tdietterich/status/2055055541542760694>

Table 1. Dataset Summary

Probe	N_{obs}	z range	Observable	Source
DESI DR2 BAO ^b	13	0.30–2.33	D_V/r_d , D_M/r_d , D_H/r_d	DESI Collaboration (2025a)
Union3	22	0.05–2.26	$\mu(z)$	Rubin et al. (2023)
Pantheon+	22 ^a	0.04–2.12	$\mu(z)$	Scolnic et al. (2022)
DES-Dovekie	19 ^a	0.04–1.14	$\mu(z)$	Popovic et al. (2025)

^aBinned to Union3 22-bin grid; Pantheon+ and DES-Dovekie have 3 empty bins at $z > 1.12$.

^bSee Table 2 for the per-tracer breakdown.

mologically relevant information lies in a space that is smooth as a function of redshift, so we bin all datasets to Union3’s common 22-bin grid for direct comparability— ~ 20 bins capture the full cosmological content. We bin using inverse-variance weighted averages. Let i index the raw (unbinned) SN points and j index the coarsened bins. With inverse-variance weights w_i and bin weights $W_j = \sum_{i \in j} w_i$, the binning matrix $B_{ji} = w_i/W_j$ propagates the raw covariance to the binned one as $C_{\text{bin}} = B C_{\text{raw}} B^T$. To ensure binned data and binned model predictions are compared consistently, we evaluate chain predictions at effective redshifts $z_{\text{eff},j}$ defined by $\mu_{\text{ref}}(z_{\text{eff},j}) = (B \mu_{\text{ref}}(z_{\text{raw}}))_j$, absorbing the bias from the nonlinearity of $\mu(z)$ across wide bins. After binning, all covariance matrices are well-behaved (positive-definite). Union3 has 22 occupied bins, Pantheon+ has 22, and DES-Dovekie has 19 (3 empty bins at $z > 1.14$). Calibration differences between datasets are absorbed by the absolute magnitude M_B marginalization (Section 2.3.2).

2.1.3. CMB Chains

We use ACT DR6 primary CMB (TT/TE/EE) + Planck 2018 low- ℓ (TT+EE) Λ CDM chains (ACT Collaboration et al. 2025), with no lensing, as our primary CMB reference, with 755,302 posterior samples from which we draw 2000 random samples for the SVD analysis.⁵ For beyond- Λ CDM extensions, we use ACT chains with additional parameters (Ω_k , A_{lens} , B_{PMF} , EDE $n = 2$) (ACT Collaboration 2025). The Planck 2018 chain (Planck Collaboration 2020) with 24,497 samples provides a cross-check (Table 5).

We also use the official Planck + DESI DR2 w_0 - w_a CDM chain from DESI Collaboration (2025b): Planck 2018 low- ℓ TT+EE + NPIPE high- ℓ CamSpec

⁵ Increasing to 5000 samples changes the normalized singular values by $< 4\%$ for the leading two modes, leaves all five V -vector inner products above 0.9999, and shifts the BAO c_0 tension by $< 0.1\sigma$.

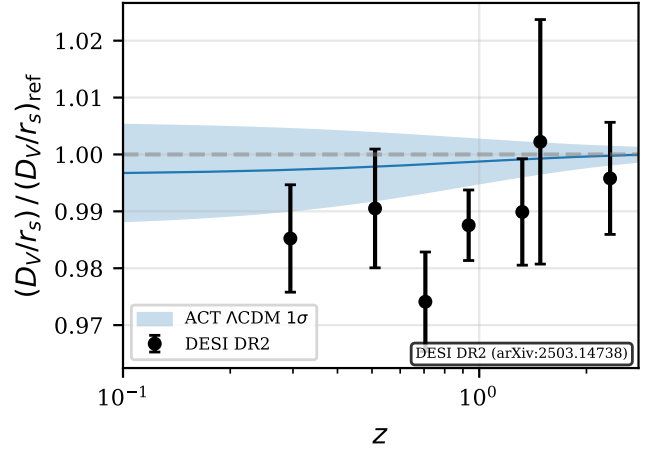


Figure 1. DESI DR2 BAO volume-averaged distance (D_V/r_d) compared to ACT Λ CDM predictions, shown as ratios relative to the Planck 2018 Λ CDM reference. The blue band shows the 1σ range from the ACT chain. DESI data show systematic deviation at intermediate redshifts ($z \sim 0.5$ – 1.5). Note: D_V is derived from D_M and D_H at each redshift for visualization only; the SVD analysis uses all 13 observables (D_M/r_d , D_H/r_d , D_V/r_d) separately (Table 2).

TTTEEE + DESI DR2 BAO, with no lensing, with $\sim 91,000$ posterior samples from which we draw 2000. This chain carries the $\sim 3\sigma$ dark energy preference that we discuss in Section 5.

2.1.4. Reference Cosmology

All distance ratios and residuals are computed relative to the Planck 2018 Λ CDM best-fit: $H_0 = 67.36$ km/s/Mpc, $\Omega_b h^2 = 0.02237$, $\Omega_c h^2 = 0.1200$ ($\Omega_m h^2 = 0.1424$), $r_d = 147.09$ Mpc.

2.2. Motivation: Distance Ratio Plots

Before developing the SVD framework, we preview the raw data-vs-model comparison to motivate the analysis (Figures 1–3).

Key observations: DESI D_V/r_d measurements lie systematically below the CMB Λ CDM prediction at $z \sim$

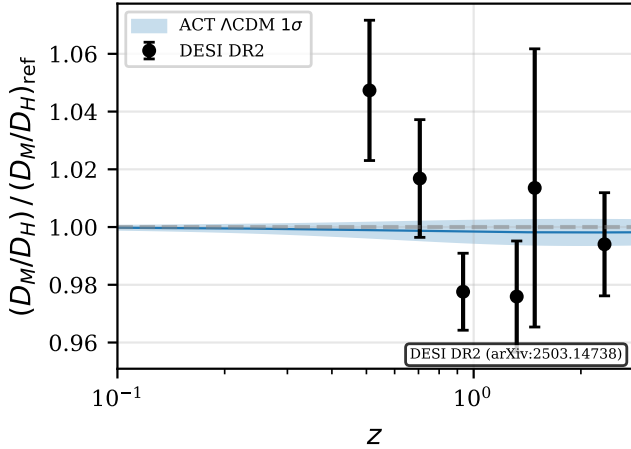


Figure 2. Anisotropic distance ratio (D_M/D_H) for DESI DR2 BAO compared to ACT Λ CDM predictions, relative to the Planck reference. The blue band shows the 1σ ACT range. This ratio is insensitive to the overall distance scale (both D_M and D_H share the same r_d calibration). The lowest-redshift point deviates from the low- z limit, the feature that would force a rapid evolution of $w(z)$ in a fit.

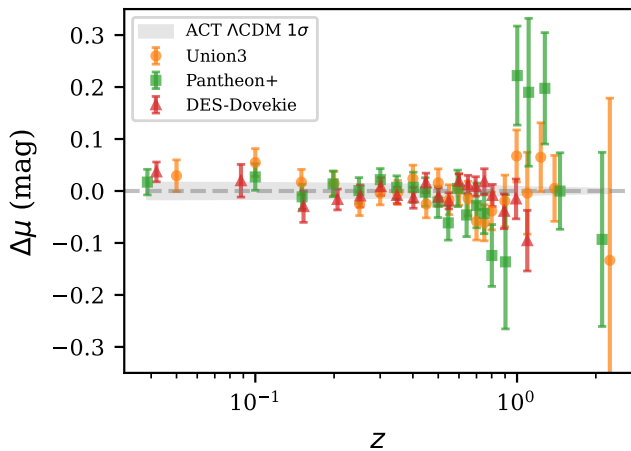


Figure 3. All three SN datasets (mean-subtracted, binned to Union3 grid) vs. the ACT Λ CDM 1σ prediction envelope. Red circles: Union3 (22 bins). Blue squares: Pantheon+ (22 bins). Green triangles: DES-Dovekie (19 bins). The ACT band is narrow ($\Delta\mu \lesssim 0.01$ mag) because ACT tightly constrains $\Omega_m h^2$.

0.5–1.5. All ratio curves converge to unity at high z because θ_* is so well measured by the CMB. The SVD analysis (Section 2.3) will formalize what these plots show qualitatively.

These data overviews (Figures 1–3) already reveal the key features that the SVD analysis will quantify. First, DESI D_V/r_d measurements lie systematically below the CMB Λ CDM prediction at $z \sim 0.5$ –1.0: the Hubble rate must be higher than the reference cosmology at

low redshifts to get the shorter distances. But θ_* fixes the distance to the last-scattering surface, so $H(z)$ must eventually drop below the reference at higher redshift to compensate—a requirement that, within the dark energy framework, implies $w < -1$ at some epoch. This is so because within the dark energy framework, the dark energy density must increase with time relative to the cosmological constant in Λ CDM, forcing the model to cross the phantom divide. Of course if we think of the combined dark-matter dark-energy fluid, whose equation of state is not close to -1 , no such requirement exists.

Second, we also plot the ratio $(D_M/D_H)/(D_M/D_H)_{\text{ref}}$ (Figure 2), which approaches unity at low z for any cosmology where the Hubble parameter can be considered constant in that limit. In that case, both terms in the ratio reduce to z , so deviations from unity at low z are small. The lowest DESI data point at $D_M/D_H = 1.04$ deviates from this limit, forcing a rapid evolution of $w(z)$ if one wants to fit this point.

Third, the SN plots show that they have substantially less constraining power than BAO: the error bars on the SN residuals are visibly wider than the 1σ ACT Λ CDM band.

2.3. SVD Methodology

Throughout this section, $s = 1, \dots, N$ indexes chain samples (we use $N = 2000$), $i = 1, \dots, D$ indexes data observables (so $D = 13$ for BAO, 22 for Union3 and Pantheon+, 19 for DES-Dovekie), and $\alpha = 0, 1, \dots, D - 1$ indexes SVD modes ordered by decreasing singular value.

We adopt one convention throughout. We call the V_α (the columns of V) *modes* or *directions*: unit vectors in whitened observable space. The scalar projection of a chain sample onto mode α is its *chain amplitude* $c_\alpha^{(s)}$, and the projection of the data onto the same direction is the *data amplitude* a_α . When we write $c_0, c_1, \dots$ without a sample superscript we mean these amplitudes (the data amplitude when comparing to data, the chain mean when comparing to the model); the directions themselves are always written $V_0, V_1, \dots$

2.3.1. BAO: Ratio Space

For each CMB chain sample, we compute the 13 BAO observables listed in Table 2 at DESI redshifts, using that sample’s own cosmological parameters—including its sample-specific sound horizon r_d . We form ratios of each chain prediction relative to the reference cosmology: $\delta_i^{(s)} = \text{pred}_i^{(s)}/\text{ref}_i - 1$, where $\text{pred}_i^{(s)}$ is the predicted observable for chain sample s and ref_i uses the Planck best-fit parameters (Section 2.1.4). The DESI

Table 2. The 13 DESI DR2 BAO observables used in the SVD analysis. Each anisotropic redshift bin contributes two observables (D_M/r_d and D_H/r_d); the lowest bin provides only D_V/r_d .

z_{eff}	Tracer	Observables	N_{obs}
0.295	BGS	D_V/r_d	1
0.510	LRG1	$D_M/r_d, D_H/r_d$	2
0.706	LRG2	$D_M/r_d, D_H/r_d$	2
0.934	LRG3+ELG1	$D_M/r_d, D_H/r_d$	2
1.321	ELG2	$D_M/r_d, D_H/r_d$	2
1.484	QSO	$D_M/r_d, D_H/r_d$	2
2.330	Ly α	$D_M/r_d, D_H/r_d$	2

observational covariance is transformed to ratio space: $C_{\text{ratio}}(i, j) = C_{\text{abs}}(i, j) / (\text{ref}_i \times \text{ref}_j)$.

2.3.2. SN: Difference Space with M_B Marginalization

For each chain sample, we compute distance modulus residuals: $\Delta\mu_i = \mu_{\text{model}}(z_i) - \mu_{\text{ref}}(z_i)$. The unknown absolute magnitude M_B is marginalized by inflating the covariance:

$$C_{\text{marg}} = C_{\text{bin}} + \sigma_M^2 \cdot \mathbf{1}\mathbf{1}^T \quad (1)$$

with $\sigma_M = 100$ (in magnitudes, vastly larger than any plausible M_B uncertainty; in the $\sigma_M \rightarrow \infty$ limit this is mathematically equivalent to projecting out the constant-offset direction, but a finite value keeps the covariance well-conditioned), adding one very large eigenvalue ($\approx N\sigma_M^2$) in the constant-offset direction. After whitening, this direction gets weight $\sim 1/\sqrt{N\sigma_M^2} \approx 0.002$ for $N = 22$ —effectively zero. The result is stable for $\sigma_M \geq 5$.

2.3.3. Whitening

We whiten using the eigendecomposition of the (ratio-transformed or M_B -marginalized) covariance: $\tilde{y} = C^{-1/2}y$. Whitening puts all directions on equal footing: in $\tilde{y}$ -space every linearly independent combination of the observables has unit measurement variance, so the subsequent SVD picks out directions of largest *model* variation across the CMB posterior, not directions that happen to have large measurement errors. In the whitened space, every direction has unit measurement variance. For SN datasets, the M_B marginalization (Eq. 1) inflates one eigenvalue to $\sim N\sigma_M^2 \approx 2.2 \times 10^5$; whitening assigns that direction negligible weight, effectively marginalizing M_B without discarding the dimension. All covariances are well-conditioned after binning, so all dimen-

sions are retained: 13 for BAO, 22 for Union3 and Pantheon+, and 19 for DES-Dovekie (whose recalibrated covariance (Popovic et al. 2025) is well-conditioned).

2.3.4. SVD

The $N \times D$ matrix of whitened chain predictions ($N = 2000$ samples, D the data dimension) is decomposed as $M = USV^T$. We do *not* mean-center M before the SVD. The offset of the chain centroid from the reference cosmology is itself a physical signal — it carries the difference between the reference cosmology’s predictions and what the CMB chain prefers, and discarding it could discard the very tension we are trying to measure. For BAO, $\langle \tilde{y} \rangle$ is therefore generically nonzero; for SN, the M_B marginalization already absorbs any constant-offset direction (Section 2.3.2), so the distinction is immaterial. The columns of V are the principal directions ordered by singular value S_α .

Chain coefficients: $c_\alpha^{(s)} = U_{s\alpha} S_\alpha$ gives the projection of chain sample s onto mode α . The chain mean $\langle c_\alpha \rangle$ and standard deviation σ_{c_α} over the N samples characterize the CMB posterior’s location and spread in direction α . Data projection: $a_\alpha = (\tilde{y}_{\text{data}}) \cdot V_\alpha$. Because the V_α are orthonormal and the data are whitened, the measurement error on a_α is 1 in whitened units for every α — every mode carries the same measurement noise by construction.

The relevant quantity is the *standard deviation*:

$$\sigma_{c_\alpha} \equiv \text{std}(c_\alpha) \quad (2)$$

which measures how much the model prediction moves across the CMB posterior in direction α , in units of measurement error. This is what we report throughout the paper. The interpretation is simple:

- $\sigma_{c_\alpha} > 1$: the low- z probe constrains direction α more tightly than the CMB. New information flows from the probe to the CMB.
- $\sigma_{c_\alpha} < 1$: the CMB already constrains direction α better than the probe can measure it.

The key point is that the SVD modes and their singular values are determined entirely by the CMB chains and the low- z covariance matrix—*before looking at the low- z data*. The data only enter when we project onto these predetermined directions.

2.3.5. Tension Formula

The tension between data and model in mode α is defined below. The numerator measures how far the data sits from the chain mean in mode α ; the denominator combines, in quadrature, the unit measurement variance

(from whitening) and the chain variance $\sigma_{c_\alpha}^2$ (the model uncertainty across the CMB posterior). Both variances are in the same whitened units.

$$\text{tension}_\alpha = \frac{a_\alpha - \langle c_\alpha \rangle}{\sqrt{1 + \sigma_{c_\alpha}^2}} \quad (3)$$

where a_α is the data projection, $\langle c_\alpha \rangle$ is the chain mean, and σ_{c_α} is the chain standard deviation (Eq. 2). With this convention, positive tension means the data amplitude a_α lies above the chain mean $\langle c_\alpha \rangle$. The “1” in the denominator is the unit measurement variance in whitened space (by construction). Thus the denominator combines two independent uncertainties: observational measurement error (always 1 in whitened units) and model uncertainty from the CMB posterior ($\sigma_{c_\alpha}^2$). This formula assumes Gaussian distributions for the c_α coefficients; Kolmogorov–Smirnov tests confirm that the leading-mode distributions are consistent with Gaussian for all four probes.

2.3.6. Parameter Formulas via β -vectors

To identify the physical meaning of each SVD mode, we regress the sigma-normalized SVD coefficient against sigma-normalized cosmological parameters:

$$\frac{c_\alpha - \langle c_\alpha \rangle}{\sigma_{c_\alpha}} = \sum_p \beta_p^\alpha \frac{X_p - \langle X_p \rangle}{\sigma_{X_p}} \quad (4)$$

where $X_p \in \{\Omega_m h^2, \Omega_b h^2, \theta_\star\}$. Because both sides are standardized, the β coefficients reflect the product of physical sensitivity and CMB posterior width: a parameter that the CMB constrains loosely will have a larger $|\beta|$ for a given physical sensitivity. The β -vector coefficients reveal which combination of CMB parameters drive each mode.

2.3.7. Extension SVD

When additional parameters are varied (e.g., w_0 – w_a , Ω_k , A_{lens}), the model predictions sweep a larger volume of observable space. The key question is whether this additional freedom opens *new measurable directions* beyond c_0 , or whether it merely expands the range of motion along the same direction.

To search for genuinely new directions, that are a priori reasonable given the model, we project out c_0 from the whitened chain predictions and decompose the residuals:

$$y_\perp = y - (y \cdot V_0) V_0 \quad (5)$$

We then perform SVD on $y_\perp$ to obtain residual modes. The correct measurability criterion is $\sigma_{\text{res}} \equiv \text{std}(c_{\text{res}})$, the standard deviation of the leading residual coefficient across the chain posterior. This measures how much the

Table 3. Leading Two SVD Mode Standard Deviations σ_{c_α} by Probe

Mode	BAO	Union3	Pantheon+	DES-Dovekie
σ_{c_0}	1.56	0.28	0.30	0.33
σ_{c_1}	0.063	0.0010	0.0011	0.0009
$\sigma_{c_0}/\sigma_{c_1}$	25	273	275	355

model prediction moves in the new direction, in units of measurement error. As a rule of thumb, we will require $\sigma_{\text{res}} > 0.3$ for a direction to be considered measurable: below this threshold, the model displacement is less than 1/3 of a measurement σ , contributing < 0.1 to χ^2 in the new direction and making the mode indistinguishable from noise. Of course this is an arbitrary threshold only used to guide our intuition.

3. THE UNIVERSAL MODE

The SVD of whitened chain predictions reveals that one mode dominates for every probe. Table 3 summarizes the per-mode standard deviations σ_{c_α} (Eq. 2).

Within Λ CDM, only the amplitude of the leading mode is measurable for any of the four probes. BAO has $\sigma_{c_0} = 1.56 > 1$: the CMB chain populates the V_0 direction by more than one BAO measurement error, so projecting the BAO data onto V_0 constrains $\Omega_m h^2$ more tightly than the CMB itself. The three SN datasets have $\sigma_{c_0} \approx 0.28$ – $0.33 < 1$: the CMB already constrains V_0 tighter than the SN data can — projecting the SN data onto V_0 adds little information about $\Omega_m h^2$ beyond the CMB prior. No probe has a second measurable Λ CDM direction ($\sigma_{c_1} \ll 1$ for all, including BAO at $\sigma_{c_1} = 0.06$).

3.1. Universality and Physical Interpretation

All four probes define almost the same c_0 direction in parameter space. Table 4 shows the β -vector coefficients (Eq. 4) for each probe:

To compare what the four probes’ leading directions V_0 have in common, we must first express them in a common space. Each probe’s V_0 is a unit vector of length D_{probe} in its own whitened observable space (e.g., 13-D for BAO, 22-D for Union3). We map each V_0 into a common 3-D CMB parameter space via the β -vector regression (Eq. 4): the unit-norm vector $\hat{\beta} = (\beta_{\Omega_m h^2}, \beta_{\Omega_b h^2}, \beta_{\theta_\star}) / \|\beta\|$ identifies the linear combination of CMB parameters that each probe’s V_0 is sensitive to. The pairwise dot product $\hat{\beta}^{(A)} \cdot \hat{\beta}^{(B)}$ is what we call the cross-probe inner product. A value of 1 means

Table 4. β -Vector Coefficients for c_0 . In all cases the three-parameter fit achieves $R^2 > 0.999$.

Probe	$\beta_{\Omega_m h^2}$	$\beta_{\Omega_b h^2}$	β_{θ_*}
BAO (DESI)	-0.90	+0.13	+0.17
Union3	-0.90	+0.18	+0.12
Pantheon+	-0.90	+0.18	+0.12
DES-Dovekie	-0.90	+0.18	+0.12

the two probes' V_0 directions correspond to identical parameter combinations; 0 would mean orthogonal. The cross-probe inner product for c_0 exceeds 0.996 for every pair. The three SN datasets define virtually identical c_0 directions; BAO differs slightly due to its sensitivity to the sound horizon r_d , but the BAO–SN inner product still exceeds 0.996. This universality is not guaranteed a priori: BAO measures D/r_d ratios while SN measures distance moduli $\mu(z)$ — at low redshift these depend on different combinations of cosmological parameters. The convergence to the same direction is a consequence of the CMB Λ CDM posterior: once the CMB pins down the parameters the residual freedom in low-redshift distance predictions runs almost entirely along one direction, which we loosely call the $\Omega_m h^2$ direction, regardless of which probe is doing the measuring.

The dominance of $\Omega_m h^2$ indicates that c_0 is primarily an $\Omega_m h^2$ proxy. Here R^2 is the fraction of the chain-sample variance of c_0 explained by a linear regression on the listed CMB parameters; $R^2 = 1$ would mean c_0 is exactly a linear combination of those parameters across the CMB posterior. Sequential multi-parameter fits quantify this:

- $\Omega_m h^2$ alone: $R^2 = 0.95$
- $\Omega_m h^2 + \Omega_b h^2$: $R^2 = 0.974$
- $\Omega_m h^2 + \Omega_b h^2 + \theta_*$: $R^2 = 0.99998$

The first line is from a practical perspective the non-trivial content: $\Omega_m h^2$ alone explains 95% of the chain variance in c_0 . Adding $\Omega_b h^2$ and θ_* absorbs the residual 5%, giving $R^2 = 0.99998$. The three-parameter $R^2 \approx 1$ is expected — the remaining Λ CDM parameters (optical depth, primordial amplitude and tilt) do not enter low-redshift distances at the linear level — but the 0.95 from $\Omega_m h^2$ alone is what makes c_0 usable as an $\Omega_m h^2$ proxy. For brevity we will refer to c_0 as the $\Omega_m h^2$ mode throughout, with the understanding that it is the specific linear combination of $(\Omega_m h^2, \Omega_b h^2, \theta_*)$ along the CMB degeneracy direction.

The sigma-normalized formula for BAO is:

$$\frac{c_0 - \langle c_0 \rangle}{\sigma_{c_0}} = -0.90 \frac{\Omega_m h^2 - \langle \Omega_m h^2 \rangle}{\sigma_{\Omega_m h^2}} + 0.13 \frac{\Omega_b h^2 - \langle \Omega_b h^2 \rangle}{\sigma_{\Omega_b h^2}} + 0.17 \frac{\theta_* - \langle \theta_* \rangle}{\sigma_{\theta_*}} \quad (6)$$

The SN β -vectors are nearly identical (Table 4), so the sigma-normalized c_0 is the same linear combination of cosmological parameters for every probe.

The difference between probes lies only in the normalization: $c_0 = 1$ corresponds to 1σ of measurement error for that probe, and since BAO and SN have very different measurement precisions, a unit of BAO c_0 and a unit of SN c_0 correspond to different shifts in $\Omega_m h^2$. Specifically, because the β -vectors are almost proportional, the c_0 values for any two probes are related by:

$$c_0^{\text{SN}} \approx \frac{\sigma_{c_0}^{\text{SN}}}{\sigma_{c_0}^{\text{BAO}}} c_0^{\text{BAO}} \quad (7)$$

For DES-Dovekie, $\sigma_{c_0}^{\text{SN}}/\sigma_{c_0}^{\text{BAO}} = 0.33/1.56 \approx 0.21$: a given cosmological shift produces a $\sim 5\times$ smaller c_0 value in SN whitened units than in BAO whitened units, reflecting the lower sensitivity of SN to $\Omega_m h^2$.

The dominance of $\Omega_m h^2$ follows quantitatively from the logarithmic derivatives of low-redshift distance observables combined with the CMB parameter covariance (Appendix A). We define the *per-sigma sensitivity* of an observable to a parameter p as the fractional change in the observable per 1σ shift in p along the CMB posterior — i.e., the log derivative times the CMB fractional error. For $D_M(z)/r_d$ and $\Omega_m h^2$, this is $\sim 5\times$ larger than for $\Omega_b h^2$, the product of (i) a log derivative $\partial \ln(D_M/r_d)/\partial \ln \Omega_m h^2 \approx 0.8$ which is $\sim 3\times$ larger than $|\partial \ln(D_M/r_d)/\partial \ln \Omega_b h^2| \approx 0.3$, and (ii) a CMB fractional error on $\Omega_m h^2$ of 0.8% which is $\sim 1.6\times$ larger than the 0.5% on $\Omega_b h^2$. The large θ_* derivative ($g_{\theta_*} \approx -4$) is suppressed by the tiny fractional error (0.025%).

The correlations between parameters in the CMB posterior further amplify the $\Omega_m h^2$ dominance: when regressing on $\Omega_m h^2$ alone, the correlated parts of $\Omega_b h^2$ and θ_* are partially captured, so that $\Omega_m h^2$ alone predicts $R^2 = 0.95$. Appendix A predicts the β -coefficients in Table 4 to within 2% for $\Omega_m h^2$ and $\sim 30\%$ for $\Omega_b h^2$ from single-redshift proxy observables. The results in the appendix are presented simply to gain some intuition about the results.

For SN, the M_B marginalization removes the absolute distance scale, leaving the *shape* of the distance–redshift relation. The relevant proxy is the distance ratio $D_M(0.3)/D_M(0.5)$, which has no r_d dependence and no M_B ambiguity. Its predicted β is similar to the BAO

proxy’s (Appendix A, Table 13), explaining why the SN β -vectors in Table 4 nearly match the BAO values. The small difference ($\beta_{\Omega_b h^2} = +0.18$ for SN vs $+0.13$ for BAO) traces to the fact that BAO has an additional $\Omega_b h^2$ sensitivity through r_d that the SN distance ratio lacks.

The MEDI framework (Weiner 2026) provides the analytic counterpart to this result: it identifies $\omega_m r_d^2$ — rather than $\Omega_m h^2$ alone — as the parameter combination that acoustic-scale measurements (BAO calibrated by the CMB) are fundamentally sensitive to. Within Λ CDM the drag-epoch sound horizon r_d is nearly fixed by early-universe physics, so $\omega_m r_d^2$ is to good approximation proportional to $\Omega_m h^2$; this is why c_0 emerges as essentially an $\Omega_m h^2$ mode, and also why regressing c_0 on $\omega_m r_d^2$ across the Λ CDM chain is no more informative than regressing on $\Omega_m h^2$ alone — the chain does not exercise r_d enough to separate the two. The value of the MEDI identification is that it holds analytically, including in extended models where r_d can shift. See the introduction for a fuller discussion of the relation to previous work.

Figure 4 shows what varying c_0 looks like in observable space. The left panel displays BAO D_V/r_d ratios (normalized to the Planck reference cosmology) for five values of c_0 spanning ± 5 in BAO whitened units, which corresponds to $\pm 5/\sigma_{c_0}^{\text{BAO}} \approx \pm 3.2$ in units of the ACT Λ CDM chain spread in V_0 (Table 3); the right panel shows the corresponding SN distance modulus residuals (mean-subtracted to marginalize M_B). The c_0 values in the two panels do not correspond to the same cosmological parameters: the normalization of c_0 is set by each probe’s measurement covariance (Eq. 7), so $c_0 = 5$ represents a $\sim 5\times$ larger shift in $\Omega_m h^2$ for BAO than for SN.

4. c_0 TENSIONS: $\Omega_m h^2$ FROM ALL PROBES

Since all probes measure the same c_0 direction (Section 3), we can compare their c_0 measurements directly. The tension formula (Eq. 3) gives four independent measurements of the same quantity, although we need to remember that the c_0 normalization differs between probes. A unit shift in BAO c_0 and a unit shift in SN c_0 both correspond to 1σ of their respective measurement errors, but these translate to different shifts in $\Omega_m h^2$ (see Section 4.1).

Table 5 summarizes the results. A few points of interpretation are needed before examining the numbers:

- **Sign convention:** Positive tension means the data projection a_0 exceeds the chain mean $\langle c_0 \rangle$. Since $\beta_{\Omega_m h^2} = -0.90$, a positive c_0 shift corresponds to *lower* $\Omega_m h^2$ (larger distances).

- **What σ_{c_0} means:** This is the standard deviation of c_0 across the CMB chain posterior—it measures how much the CMB model prediction varies in the c_0 direction, in units of measurement error. It is *not* the measurement error, which is always 1 in whitened space by construction. The tension denominator $\sqrt{1 + \sigma_{c_0}^2}$ combines both.
- **σ_{c_0} and constraining power:** BAO has $\sigma_{c_0} = 1.56 > 1$. This means BAO constrains $\Omega_m h^2$ more tightly than the CMB: DESI data carries genuinely new information about $\Omega_m h^2$ beyond what the CMB already provides. The three SN datasets have $\sigma_{c_0} \approx 0.28\text{--}0.33 < 1$: the CMB already provides a tighter constraint than these SN measurements, so their c_0 tensions are measured against a prior that is already narrower than the data.

BAO prefers lower $\Omega_m h^2$ than the CMB (positive c_0 tension at $+2.2\sigma$), while the three SN datasets prefer higher $\Omega_m h^2$ —Union3 at -1.7σ , Pantheon+ at -1.1σ , and DES-Dovekie at -0.8σ . None of the SN tensions is individually significant, and because $\sigma_{c_0}^{\text{SN}} < 1$ the SN datasets do not have the constraining power on c_0 to confirm or refute the BAO result; the opposite-sign pattern is suggestive but should not be interpreted as a measured disagreement.

An important point to remember when interpreting Figure 5: the c_0 normalization differs between probes. A unit shift in BAO c_0 and a unit shift in SN c_0 both correspond to 1σ of their respective measurement errors, but these translate to different shifts in $\Omega_m h^2$ (see Section 4.1). The figure is designed to visualize *tensions*, not to compare $\Omega_m h^2$ values directly.

4.1. The $\Omega_m h^2$ Money Plot

Since c_0 is a near-perfect proxy for $\Omega_m h^2$ (Eq. 6), we can convert each probe’s c_0 measurement to an implied $\Omega_m h^2$ value. This is the value of $\Omega_m h^2$ implied by the data if one slides along the CMB posterior’s degeneracy direction in $(\Omega_m h^2, \Omega_b h^2, \theta_*)$. The conversion inverts the β -vector formula, setting $\Omega_b h^2$ and θ_* to their chain mean values:

$$\Omega_m h_{\text{probe}}^2 = \langle \Omega_m h^2 \rangle_{\text{chain}} + (a_0 - \langle c_0 \rangle) \times \frac{\sigma_{\Omega_m h^2}}{\beta_{\Omega_m h^2} \cdot \sigma_{c_0}} \quad (8)$$

where $\beta_{\Omega_m h^2}$ is the β -vector coefficient for $\Omega_m h^2$, and $\sigma_{\Omega_m h^2}, \sigma_{c_0}$ are the chain standard deviations. The measurement error on $\Omega_m h^2$ from the $\sigma = 1$ whitened-space uncertainty is $\sigma_{\Omega_m h^2}^{\text{meas}} = |\sigma_{\Omega_m h^2} / (\beta_{\Omega_m h^2} \cdot \sigma_{c_0})|$. An additional marginalization uncertainty arises from

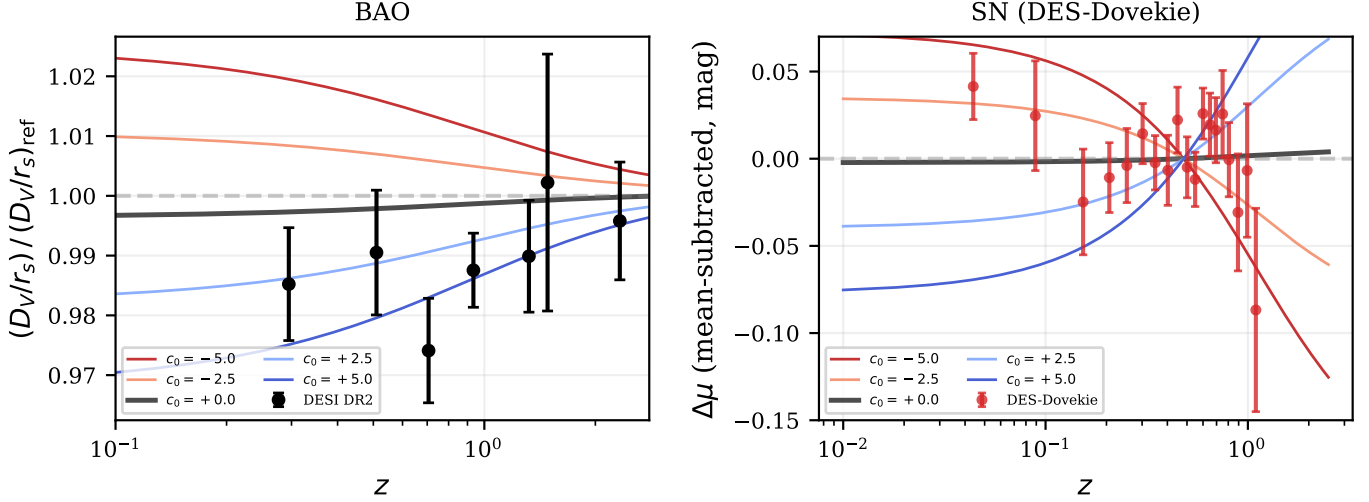


Figure 4. c_0 family of curves in observed space. *Left:* BAO D_V/r_d ratio (normalized to the Planck reference cosmology) for five values of c_0 spanning ± 5 whitened units ($\approx \pm 3\sigma$ of the ACT chain). Black points: DESI DR2. *Right:* SN distance modulus residuals $\Delta\mu$ (mean-subtracted) for the same c_0 range. Points: DES-Dovekie. The c_0 normalization differs between panels (Eq. 7): BAO $c_0 = 5$ corresponds to a $\sim 5\times$ larger $\Omega_m h^2$ shift than SN $c_0 = 5$.

NOTE—Columns: $\langle c_0 \rangle$ is the mean of the CMB chain’s c_0 distribution (the model prediction in the V_0 direction); σ_{c_0} its standard deviation across the chain (model uncertainty, in units of the measurement error); a_0 is the data amplitude, the projection of the measured low- z data onto V_0 (measurement error 1 by construction); and the tension is $(a_0 - \langle c_0 \rangle)/\sqrt{1 + \sigma_{c_0}^2}$ (Eq. 3). Values in parentheses use the Planck 2018 Λ CDM chain as the CMB reference instead of ACT DR6.

Table 5. c_0 Tension per Probe

Probe	$\langle c_0 \rangle$	σ_{c_0}	a_0	Tension
BAO (DESI)	+0.59 (−0.31)	1.56 (1.85)	+4.69	+2.2 σ (+2.4 σ)
Union3	+0.13 (−0.06)	0.28 (0.34)	−1.59	−1.7 σ (−1.4 σ)
Pantheon+	+0.14 (−0.06)	0.30 (0.36)	−0.97	−1.1 σ (−0.9 σ)
DES-Dovekie	+0.16 (−0.07)	0.33 (0.40)	−0.74	−0.8 σ (−0.6 σ)

the CMB posterior widths of $\Omega_b h^2$ and θ_* ; this adds $\sigma_{\text{marg}} \approx 0.0003$ in quadrature, a modest correction ($\sim 7\%$ for BAO, negligible for SN).

Figure 6 shows the result. BAO provides the tightest $\Omega_m h^2$ constraint of the four probes — tighter than the CMB itself — while the three SN measurements have total errors $\sim 5\times$ larger.

4.2. Physical Interpretation of the Mode Shape

The c_0 mode measures $\Omega_m h^2$, and the physical effect of varying $\Omega_m h^2$ at fixed θ_* is a characteristic tilt of the Hubble rate $H(z)$ (Figure 7). Increasing $\Omega_m h^2$ raises $H(z)$ at high redshift (where $H \propto \sqrt{\Omega_m h^2}$ in the matter era), but the θ_* constraint forces H_0 to *decrease* to compensate (keeping $r_*/D_A(z_{\text{rec}})$ fixed). The result is a crossover near $z \sim 1\text{--}2$: higher $\Omega_m h^2$ means faster expansion at $z > 2$ but slower at $z < 1$. This crossover

falls squarely in the DESI BAO redshift range, explaining why BAO is sensitive to $\Omega_m h^2$.

For distances, lower $\Omega_m h^2$ (positive c_0) produces smaller D_V/r_d ratios at all BAO redshifts (Figure 4, left panel): the slower high- z expansion reduces distances, while the higher H_0 reduces r_d -normalized quantities further. For SN, the same physics manifests as a tilt in $\mu(z)$: positive c_0 shifts distance moduli downward (brighter supernovae) at low z and upward at high z . Note that when making these plots we just varied $\Omega_m h^2$ while keeping $\Omega_b h^2$ and θ_* fixed for simplicity. Varying $\Omega_b h^2$ and θ_* along the degeneracy does not change things qualitatively.

4.3. Goodness of Fit and Robustness

Beyond the c_0 tension, a natural question is whether the remaining SVD modes are well-behaved. We com-

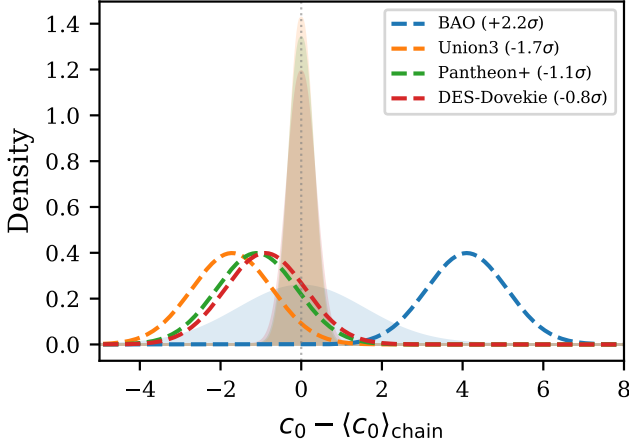


Figure 5. Per-probe c_0 distributions. *Shaded regions:* CMB chain predictions (model uncertainty in each probe’s c_0). *Dashed curves:* measurement likelihoods centered at the data c_0 value ($\sigma = 1$ in whitened space by construction). The BAO chain distribution (shaded, wide) has $\sigma_{c_0} = 1.56$, wider than the measurement likelihood, reflecting BAO’s ability to constrain $\Omega_m h^2$ beyond the CMB prior. The three SN chain distributions (shaded, narrow, $\sigma_{c_0} \approx 0.28\text{--}0.33$) are much narrower than their measurement likelihoods, indicating SN data adds little constraining power on c_0 .

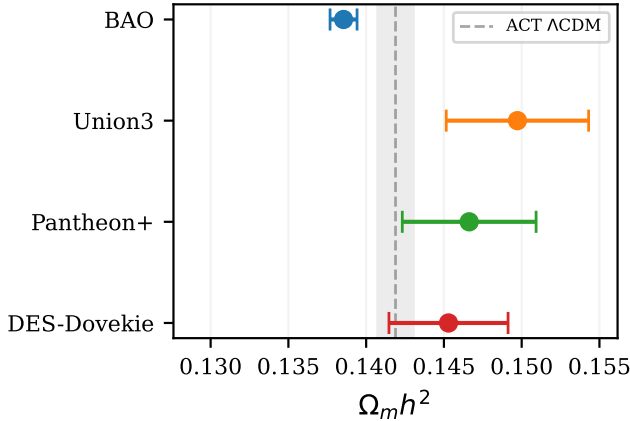


Figure 6. Effective $\Omega_m h^2$ measurements from all four distance probes, derived from c_0 via the β -vector inversion (Eq. 8). Error bars show total uncertainty (measurement + marginalization over $\Omega_b h^2$ and θ_* , added in quadrature). The gray band shows the ACT Λ CDM posterior width. BAO provides a tighter constraint than the CMB.

pute a dimension-limited χ^2 :

$$\chi^2(K) = \sum_{\alpha=0}^{K-1} \frac{(a_\alpha - \langle c_\alpha \rangle)^2}{1 + \sigma_{c_\alpha}^2} \quad (9)$$

For BAO, $\chi_{13}^2 = 15.7$ ($p = 0.27$); excluding c_0 , $\chi_{12}^2 = 10.8$ ($p = 0.55$). The remaining 12 modes are well-behaved. All three SN datasets also show good fits:

Union3 $\chi_{22}^2 = 26.8$ ($p = 0.22$), Pantheon+ $\chi_{22}^2 = 28.4$ ($p = 0.16$), DES-Dovekie $\chi_{19}^2 = 20.0$ ($p = 0.39$). Excluding c_0 leaves Union3 $\chi_{21}^2 = 24.0$ ($p = 0.29$), Pantheon+ $\chi_{21}^2 = 27.3$ ($p = 0.16$), DES-Dovekie $\chi_{18}^2 = 19.3$ ($p = 0.37$). No dataset shows anomalous higher-mode structure. Of course this is a very crude test, as it could be that a significant discrepancy in one specific direction would get diluted in this statistic. Thus in the next sections we will perform a more detailed analysis of the higher-mode structure using specific physical models as guidance to identify promising directions based on our theoretical expectations.

We also verify that the tensions are robust to the redshift range: restricting all datasets to $z \leq 0.9$ changes tensions modestly (BAO: $+2.2\sigma \rightarrow +1.9\sigma$; Union3: $-1.7\sigma \rightarrow -2.0\sigma$; Pantheon+: $-1.1\sigma \rightarrow -1.8\sigma$; DES-Dovekie: $-0.8\sigma \rightarrow -0.6\sigma$). The high-redshift bins do not drive the tensions.

4.4. Summary Within Λ CDM

Within Λ CDM, there is a tension between DESI and the CMB which lives in the direction we are calling the $\Omega_m h^2$ direction—the worst-constrained CMB combination parameter for low- z expansion history. No other direction is measurable: $\sigma_{c_1} = 0.06$ for BAO, ≈ 0.001 for SN (Table 3). This fact is not particularly surprising. The BAO c_0 tension is mild but real: BAO and the CMB prefer different $\Omega_m h^2$ values at $+2.2\sigma$. The three SN datasets do not have enough constraining power in V_0 ($\sigma_{c_0}^{\text{SN}} < 1$) to either confirm or refute the BAO signal at its precision; their tensions are mild or null but cannot independently corroborate a true cosmological shift. In the next section, we extend the analysis to $w_0\text{--}w_a$ dark energy and ask whether this tension opens genuinely new directions, or whether the DESI $w_0\text{--}w_a$ preference reduces primarily to the c_0 tension once the data are decomposed in the Λ CDM SVD basis.

5. DYNAMICAL DARK ENERGY

The previous section established that within Λ CDM, the only measurable direction is c_0 (which we loosely call $\Omega_m h^2$). We now ask: when the dark energy equation of state is freed from $w = -1$, does the tension migrate into genuinely new directions—or is it simply c_0 tension repackaged in a $w_0\text{--}w_a$ framework?

5.1. Methodology

To answer this question, we extend the SVD analysis by varying dark energy parameters. We fix the matter content to the Planck Λ CDM best-fit ($\Omega_m h^2 = 0.1424$, $\Omega_b h^2 = 0.02237$) and compute BAO and SN observables on a 100×100 grid spanning $w_0 \in [-1.5, 0]$ and

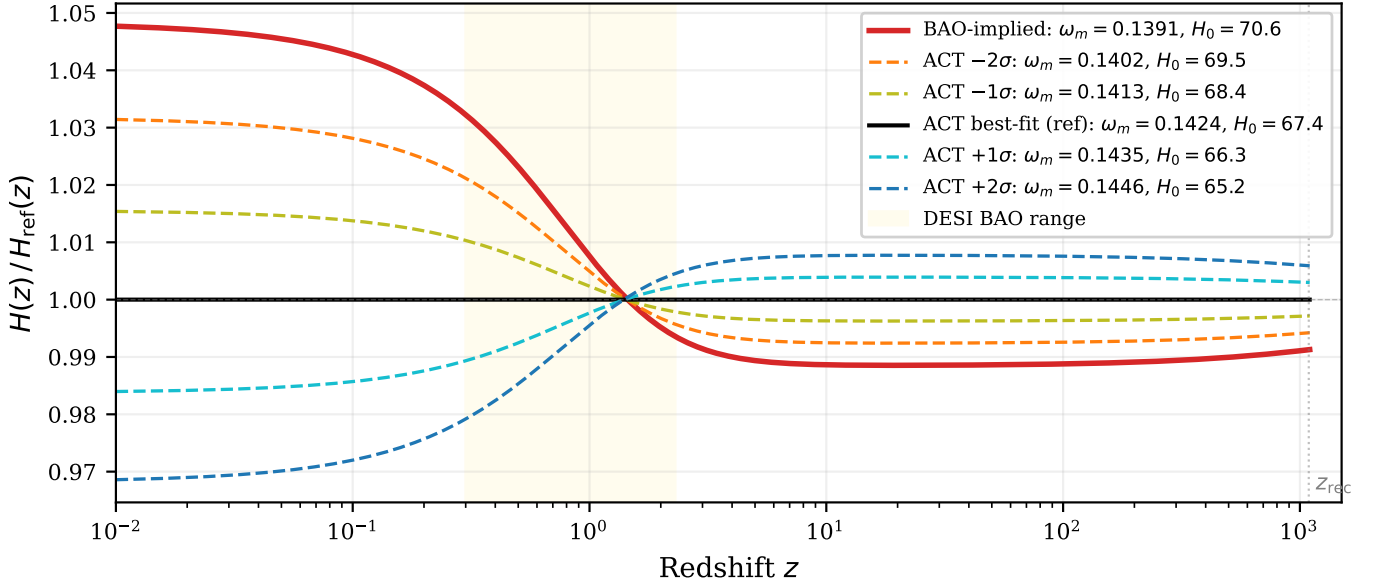


Figure 7. Hubble rate ratio $H(z)/H_{\text{ref}}(z)$ for varying $\Omega_m h^2$ at fixed θ_* (flat Λ CDM). Higher $\Omega_m h^2$ (blue, dashed) raises H at high z but lowers H_0 ; the crossover near $z \sim 2$ falls within the DESI BAO range (shaded). The red curve shows the BAO-implied $\Omega_m h^2$ from the c_0 measurement ($\Omega_m h^2 = 0.139$, $H_0 = 70.6$).

$w_a \in [-3, 1.5]$. This grid maps out how each observable responds to dark energy. In doing so, we keep $\Omega_k = 0$ and θ_* fixed. Note that we could have varied $\Omega_m h^2$ and reached the same conclusions because we will keep the V_0 direction fixed to what we found in Section 3.

Crucially, we preserve the Λ CDM c_0 direction: we set $V_0 = V_0^{\Lambda\text{CDM}}$ (the $\Omega_m h^2$ direction from Section 3) and project it out of the grid predictions before performing SVD on the residuals. The resulting modes form an orthonormal basis where:

- $V_0 (= c_0)$: identical to the Λ CDM $\Omega_m h^2$ direction, unchanged.
- $V_1 (= c_1)$: the first dark energy direction orthogonal to $\Omega_m h^2$ —the direction in which (w_0, w_a) most affects the observables.
- $V_2, V_3, \dots$: not additional dark-energy degrees of freedom — w_0-w_a has only two — but nonlinear residuals of the $(w_0, w_a) \rightarrow$ observables map. If the distance predictions were linear in (w_0, w_a) , only V_1 would have non-trivial grid range; the presence of structure in V_2, V_3 measures how much the distance map curves across the grid.

We fix V_0 to the Λ CDM result of Section 3 rather than re-running the SVD freely. Without this projection, the w_0-w_a grid’s leading mode would partly re-discover $\Omega_m h^2$ -like motion (because varying (w_0, w_a) shifts distances in a way that overlaps strongly with shifting $\Omega_m h^2$), and we would learn nothing new about dark

energy. Projecting out V_0 isolates the directions that w_0-w_a freedom genuinely opens *beyond* what Λ CDM already populates.

We then project both the low- z data and the Planck+DESI w_0-w_a chain posterior onto this basis. The Planck+DESI chain is a joint Planck 2018+DESI DR2 analysis in flat w_0-w_a CDM, produced by the DESI collaboration (DESI Collaboration 2025b) (Section 2.1); it is the chain whose $\sim 3\sigma$ dark energy preference we aim to reinterpret.

5.2. Two Modes Become Measurable

Adding two parameters (w_0, w_a) could open two new measurable directions, at least if we think of things linearly. But c_0 absorbs one slot: the most measurable direction in the w_0-w_a extension coincides with the Λ CDM c_0 (the $\Omega_m h^2$ direction). This leaves c_1 as the first genuinely new dark-energy-only direction.

To quantify measurability, we examine the *grid range*—the total variation of each SVD coefficient c_α across the (w_0, w_a) grid, in units of measurement error. Table 6 summarizes the results for all probes.

For BAO, the c_1 grid range (34) is $\sim 3\times$ smaller than c_0 (104), with $c_1/c_0 = 0.33$ —dark energy has substantially less leverage on c_1 than on the $\Omega_m h^2$ direction. The range is however much larger than one and thus we can expect to see a significant signal there if the dynamical dark energy is the correct origin of the tension. For SN, the pattern is similar: c_1 range ($\sim 8-9$) is smaller

Table 6. w_0 - w_a Grid Ranges by Probe

Probe	c_0 range	c_1 range	c_2 range	c_3 range	c_1/c_0
BAO	104	34	6.7	3.8	0.33
Union3	45	9.4	3.0	< 1	0.21
Pantheon+	50	9.4	2.7	< 1	0.19
DES-Dovekie	55	8.3	2.7	< 1	0.15

NOTE—Grid ranges: $\max(c_\alpha) - \min(c_\alpha)$ over the (w_0, w_a) grid, in units of measurement error. Each probe uses its own SVD basis (V_0 from Λ CDM chain, $V_1 \dots$ from w_0 - w_a grid residuals). In w_0 - w_a CDM, c_2 and higher are deterministic functions of c_0 and c_1 (only two free DE parameters), so these ranges are not independent. BAO c_4 and c_5 ranges are < 1 (unmeasurable). SN c_0 ranges differ across datasets because whitening normalizes by each probe’s measurement errors.

than c_0 (~ 45 – 55), with $c_1/c_0 \approx 0.15$ – 0.21 . The range is also larger than one.

For BAO, two further directions (V_2, V_3) carry non-trivial grid range (Table 6). With only two free parameters (w_0, w_a), a linear response would populate exactly two SVD directions; the extra grid range in V_2, V_3 reflects nonlinearities in the $(w_0, w_a) \rightarrow$ distance map, not new degrees of freedom.

This should be contrasted with the Λ CDM situation, where $\sigma_{c_1} = 0.06$ (BAO) and ≈ 0.002 (SN)—Table 3. (In Λ CDM all D SVD directions exist, but beyond c_0 they all have $\sigma_{c_\alpha} \ll 1$ and are individually unmeasurable; they are not marginalized away but simply carry no signal, as the goodness-of-fit test in Section 4.3 confirms.) In the w_0 - w_a extension the first new direction V_1 is measurable: its data amplitude a_1 — which we will call c_1 — becomes a genuine constraint on the dark-energy parameter space, no longer noise as it was in Λ CDM.

5.3. w_0 - w_a Contours and Dark Energy Sensitivity

Figure 8 shows contours of constant c_0 and c_1 in the (w_0, w_a) plane for BAO and Union3. Several features are immediately apparent:

- **Contours are straight:** Over most of the parameter space the contours are relatively straight, which means each of these variables measures the equation of state at a specific pivot redshift.
- **The spacing of the contours differs:** The spacing encodes how measurable the parameter is and is another way of seeing the information encoded in the ranges of c_0 and c_1 (Table 6). It is clear

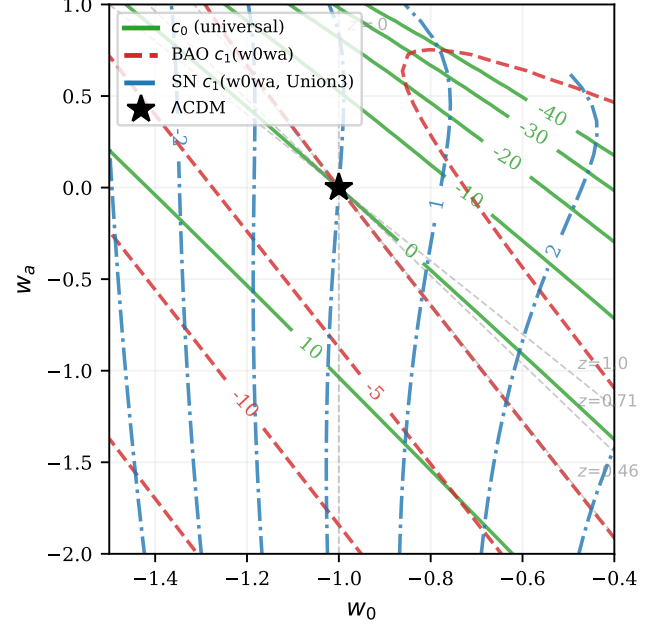


Figure 8. SVD mode contours in the w_0 - w_a plane. Green solid: constant- c_0 contours (diagonal—probes w at $z \approx 0.75$), spaced by $\Delta c_0 = 10$. Red dashed: BAO c_1 contours (diagonal—probes w at $z \approx 0.5$), spaced by $\Delta c_1 = 5$. Blue dash-dot: SN/Union3 c_1 contours (nearly vertical—probes $\sim$ pure w_0), spaced by $\Delta c_1 = 1$. All spacings are in whitened (measurement-error) units, so the much tighter c_0 spacing directly reflects its greater measurability (cf. the grid ranges in Table 6). Black star: Λ CDM point ($w_0 = -1, w_a = 0$). Grid computed with fixed Planck Λ CDM matter parameters. Pantheon+ and DES-Dovekie c_1 contours have the same orientation as Union3, with normalization factors of 1.13 and 1.01 respectively (pairwise correlations > 0.999).

that c_0 is much more measurable than c_1 , and that BAO has more discriminatory power than SN.

- **The SN c_1 contours are nearly vertical:** SN are measured at low redshift where $w(z) \approx w_0$, so SN c_1 is nearly pure w_0 sensitivity. The SN c_1 pivot redshift is $z \approx 0$ (Section 5.6).
- **BAO and SN c_1 contours are oriented in very different directions:** The BAO c_1 contours run diagonally while the SN c_1 contours are nearly vertical. This is why combining BAO and SN breaks the w_0 - w_a degeneracy.

The c_1 directions for the three SN datasets are nearly proportional: $c_1^{\text{P}+} \approx 1.13 c_1^{\text{Union3}}$ and $c_1^{\text{DES}} \approx 1.01 c_1^{\text{Union3}}$ (pairwise correlations > 0.999), so Union3 is representative in the figure. The normalization differences reflect the different measurement precisions of each dataset.

Table 7. c_1 Tensions and w_0 - w_a Chain Predictions

Probe	a_1	c_1 tension	$\langle c_1 \rangle_{\text{chain}}$	$\sigma_{c_1}^{\text{chain}}$
BAO	+1.24	+1.2 σ	+0.71	0.89
Union3	-1.49	-1.5 σ	-3.01	1.08
Pantheon+	+0.23	+0.2 σ	-3.41	1.21
DES-Dovekie	-0.75	-0.8 σ	-3.05	1.08

NOTE— a_1 : data projection onto the dark energy direction V_1 . Tension: vs. ACT Λ CDM ($\approx a_1$ since $\sigma_{c_1}^{\Lambda\text{CDM}} \ll 1$). Last two columns: Planck+DESI w_0 - w_a chain prediction for each probe's c_1 .

5.4. c_1 Tensions and the Dark Energy Direction

We now ask: does any probe show significant tension in the dark energy direction c_1 ? We compute tensions against the ACT Λ CDM chain, which predicts $\langle c_1 \rangle \approx 0$ with negligible spread when projected onto the w_0 - w_a dark energy direction V_1 ($\sigma_{c_1}^{\Lambda\text{CDM}} = 0.046$ for BAO, ≈ 0.001 for SN). The tension simplifies to $\text{tension}_1 \approx a_1$, where a_1 is the data projection onto V_1 .

Table 7 summarizes the results. For each probe, we also show the Planck+DESI w_0 - w_a chain prediction—this is the joint Planck 2018+DESI DR2 analysis in flat w_0 - w_a CDM (DESI Collaboration 2025b), whose $\sim 3\sigma$ dark energy preference we aim to reinterpret.

BAO c_1 tension is +1.2 σ , consistent with Λ CDM.

The dark energy direction shows only mild tension. The dominant +2.2 σ DESI tension lives in c_0 ($\Omega_m h^2$), not in the dark energy mode. Explicitly, BAO measures a data amplitude $a_1 = +1.24$ in the dark-energy direction, carrying unit measurement error in whitened units ($a_1 = +1.24 \pm 1.0$), against the Λ CDM expectation $\langle c_1 \rangle \approx 0$ ($\sigma_{c_1}^{\Lambda\text{CDM}} = 0.046$); equivalently, this corresponds to $w(z = 0.46) = -0.94 \pm 0.052$ at the c_1 pivot (Section 5.6). A 2D view of how these mode tensions map into the (w_0, w_a) plane is given in Figure 12 (Section 5.5), which overlays the Planck+DESI chain posterior with the SVD-truncated χ^2 contours.

The SN c_1 tensions are all mild (-1.5σ to $+0.2\sigma$). The Planck+DESI chain (which includes BAO but no SN) predicts $\langle c_1 \rangle \approx -3$ for SN—far from zero because the chain's preferred dark energy ($w_0 \approx -0.43$, $w_a \approx -1.69$) extrapolates to specific distance patterns at SN redshifts. All three SN datasets are displaced from this prediction, suggesting the chain's dark energy extrapolation overshoots the SN data (Figure 9 and Figure 10).

5.5. BAO Mode Structure: Where the Chain Lives

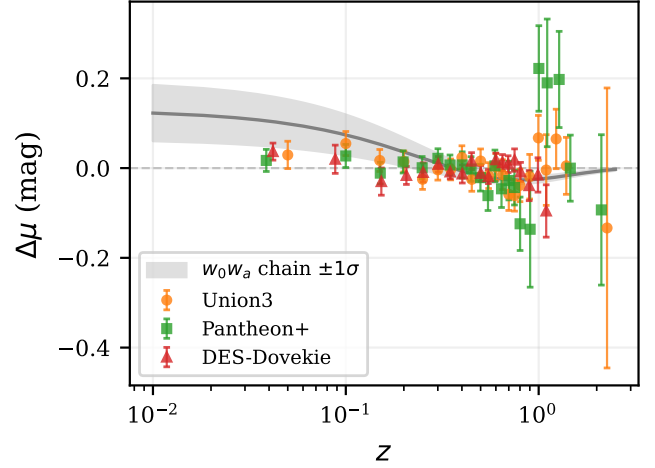


Figure 9. Planck+DESI w_0 - w_a chain SN prediction (gray band, $\pm 1\sigma$) vs. all three SN datasets. Each dataset is independently shifted by its optimal display offset M_{disp} (zero χ^2 cost due to M_B marginalization).

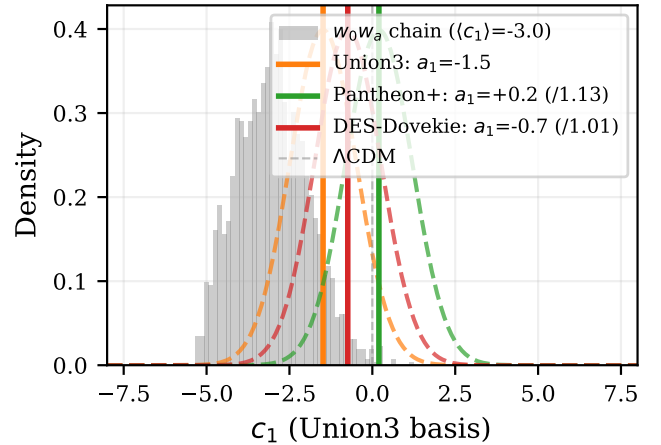


Figure 10. SN c_1 (dark energy mode). Gray histogram: Planck+DESI w_0 - w_a chain prediction ($\langle c_1 \rangle \approx -3$). Dashed Gaussians: data measurements a_1 from each SN dataset (unit measurement uncertainty). All datasets are rescaled to the Union3 c_1 basis (proportionality factors: $c_1^{\text{P}+} \approx 1.13 c_1^{\text{U}3}$, $c_1^{\text{DES}} \approx 1.01 c_1^{\text{U}3}$). All three lie much closer to $a_1 = 0$ than to the chain prediction, collectively displaced from it.

Table 8 compares the BAO data projections a_α with the Planck+DESI w_0 - w_a chain means $\langle c_\alpha \rangle$ for the first six modes.

The data amplitudes $a_1 \approx a_2 \approx a_3 \approx 1.3$ are all comparable, so no single mode dominates the data's departure from Λ CDM. The Planck+DESI w_0 - w_a chain, however, is *centered* far from Λ CDM in c_2 ($\langle c_2 \rangle = +2.00$) more than in c_1 ($\langle c_1 \rangle = +0.71$). Higher vectors are used primarily to try to accommodate the high D_M/D_H ratio of the lowest-redshift point; Figure 11

Table 8. BAO SVD Coefficients: Data vs. Planck+DESI w_0 – w_a Chain

	c_0	c_1	c_2	c_3	c_4	c_5
a_α (data)	+4.69	+1.24	+1.30	+1.32	+0.42	−0.40
$\langle c_\alpha \rangle$ (chain)	+4.48	+0.71	+2.00	+0.23	+0.00	−0.03
σ_{c_α} (chain)	0.92	0.89	0.68	0.15	0.03	0.01

shows how the characteristic D_V dip at $z \sim 0.5$ – 1.0 emerges only once c_2 and c_3 are included.

Figure 12 validates this decomposition. Panel (A) shows that a 2-mode (c_0, c_1) truncated χ^2 does not reproduce the chain posterior contours—two modes are insufficient. Panel (B) shows that the 4-mode truncation closely matches the chain, demonstrating that four modes capture the BAO information content. The cumulative $\Delta\chi^2$ values are -22.0 (c_0), -23.6 (c_0+c_1), -25.3 ($c_0+c_1+c_2$), -27.0 ($c_0+\dots+c_3$).

Combining the $+2.2\sigma$ BAO tension in c_0 (the $\Omega_m h^2$ direction, Section 4) with the $+1.2\sigma$ BAO tension in c_1 (the dark-energy direction) — and noting that by construction V_0 is the $\Omega_m h^2$ combination while V_1 is the dark-energy combination orthogonal to it — the BAO preference for w_0 – w_a over Λ CDM is driven by the matter direction, not the dark-energy direction.

5.6. Pivot Redshift Analysis

Since the contours of constant c_α are approximately straight in the (w_0, w_a) plane (Figure 8), each mode constrains the dark energy equation of state at a specific pivot redshift. We fit

$$c_\alpha = A_\alpha (1 + w_0) + B_\alpha w_a \quad (10)$$

on the (w_0, w_a) grid, weighted by the 2-mode χ^2 to focus on the region where the data lives. By construction $c_\alpha = 0$ at Λ CDM ($w_0 = -1, w_a = 0$). Since $w(z) = w_0 + w_a z/(1+z)$, Eq. 10 implies $c_\alpha = A_\alpha [w(z_p) + 1]$ at the pivot redshift $z_p = B_\alpha/(A_\alpha - B_\alpha)$. The data measurement a_α then gives $w(z_p) = a_\alpha/A_\alpha - 1$ with uncertainty $\sigma_w = 1/|A_\alpha|$; for c_0 , we add the $\Omega_m h^2$ uncertainty in quadrature (Section 4.1).

The BAO fits give:

$$c_0 \approx -24.7 (1 + w_0) - 10.5 w_a \quad (z_p = 0.74, R^2 = 0.99) \quad (11)$$

$$c_1 \approx +19.1 (1 + w_0) + 6.0 w_a \quad (z_p = 0.46, R^2 = 1.00) \quad (12)$$

Figure 13 shows $w(z_{\text{pivot}})$ from both the Planck+DESI chain (colored histograms) and the data projections

(black dashed Gaussians). At the c_0 pivot ($z = 0.74$), both chain and data show $w \approx -1.2$, deviating from Λ CDM—but this is the $\Omega_m h^2$ direction, not the independent dark energy direction. At the BAO c_1 pivot ($z = 0.46$), w is consistent with -1 from both chain and data.

At the c_1 pivot ($z = 0.46$) the constraint $w = -0.94 \pm 0.052$ is tighter than the constraint inferred at the c_0 pivot ($z = 0.74$). This is because c_1 is, by construction, the dark-energy direction orthogonal to $\Omega_m h^2$, so its measurement does not get inflated by the $\Omega_m h^2$ marginalization. Combining the two pulls the joint constraint on $w(z)$ closer to -1 : c_1 is consistent with -1 at the few per-cent level *and* has a smaller error than c_0 .

5.7. Summary

The w_0 – w_a analysis reveals a clean separation:

1. **c_0 (universal):** The $\Omega_m h^2$ tension persists unchanged, appearing as $w \neq -1$ at $z \approx 0.7$.
2. **BAO c_1 :** Mild ($+1.2\sigma$). At the DE pivot $z = 0.46$, $w = -0.94 \pm 0.052$ —consistent with a cosmological constant.
3. **BAO higher modes:** The Planck+DESI chain’s preferred $w(z)$ has a sharp low-redshift feature that the broad, smooth leading dark-energy mode V_1 cannot represent. This is why the chain sits far from Λ CDM in c_2 ($\langle c_2 \rangle = +2.0$) but barely in c_1 , even though V_1 is a priori the more measurable dark-energy direction (see Section 5.5).
4. **SN c_1 :** Mild (-1.5σ to $+0.2\sigma$). The Planck+DESI chain overshoots the SN data.

The DESI w_0 – w_a “detection” is c_0 tension repackaged. The actual genuinely new dark energy direction is consistent with a cosmological constant and the (w_0, w_a) grid contains many directions in which a real dynamical-dark-energy signal would have produced a measurable c_1 tension. None is observed.

6. SPATIAL CURVATURE

Spatial curvature modifies distances at all redshifts and is the other beyond- Λ CDM extension that opens a genuinely new measurable direction in the BAO data ($\sigma_{\text{res}} = 2.11$, exceeding the 0.3 threshold of Section 2.3.7). This makes curvature analogous to the w_0 – w_a analysis of Section 5: both c_0 and c_1 carry information about the new parameter.

6.1. Two Directions Sensitive to Ω_k

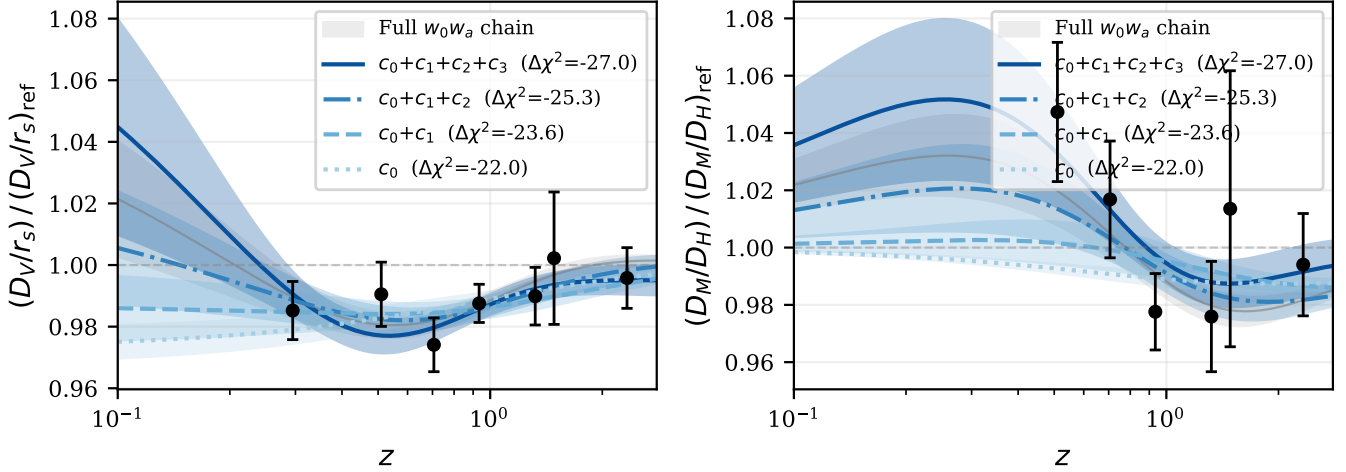


Figure 11. BAO data-centered reconstruction bands. Bands centered on data projections a_α with unit-Gaussian width per mode. Four levels: c_0 only (dotted), c_0+c_1 (dashed), $c_0+c_1+c_2$ (dash-dot), $c_0+c_1+c_2+c_3$ (solid). Gray: full w_0 - w_a chain $\pm 1\sigma$. The DV dip at $z \sim 0.5$ – 1.0 emerges when c_2 and c_3 are included.

Using the ACT chain in which Ω_k is varied, we regress the chain values of c_0 on $(\Omega_m h^2, \Omega_k)$ — the same procedure as Eq. 4, with $\Omega_b h^2$ and θ_* fixed at the chain means (they remain subdominant in the regression):

$$\frac{c_0 - \langle c_0 \rangle}{\sigma_{c_0}} = -0.17 \frac{\Omega_m h^2 - \langle \Omega_m h^2 \rangle}{\sigma_{\Omega_m h^2}} + 1.06 \frac{\Omega_k - \langle \Omega_k \rangle}{\sigma_{\Omega_k}} \quad (R^2 = 0.993) \quad (13)$$

Curvature dominates: $\beta_{\Omega_k} = +1.06$ versus $\beta_{\Omega_m h^2} = -0.17$. The reduction from $\beta_{\Omega_k} = +1.06$ in the two-parameter fit to $+0.99$ in the one-parameter fit on Ω_k alone reflects the residual $\Omega_m h^2$ - Ω_k correlation ($r = 0.42$): regressing on Ω_k alone absorbs the part of the $\Omega_m h^2$ variation that is correlated with Ω_k via the degeneracy. This is different from the Λ CDM chain ($\beta_{\Omega_m h^2} = -0.90$, with Ω_k fixed at zero). The change comes from the *geometric degeneracy*: in Λ CDM the CMB pins $\Omega_m h^2$ tightly via the acoustic peak heights, but in the open chain the CMB only pins the combination $\Omega_m h^3$ tightly (through θ_*), so the chain populates a 1-D ridge in $(\Omega_m h^2, \Omega_k)$ along which low- z distances vary primarily as a function of Ω_k .

However, these two parameters are not independent in the CMB posterior. The geometric degeneracy correlates $\Omega_m h^2$ and Ω_k ($r = 0.42$ in the chain): fitting the CMB with larger Ω_k requires a compensating shift in $\Omega_m h^2$. Consequently, c_0 along the CMB-allowed surface is effectively a function of Ω_k alone. Regressing c_0 directly on Ω_k —which accounts for the correlated $\Omega_m h^2$ variation—gives

$$\frac{c_0 - \langle c_0 \rangle}{\sigma_{c_0}} = +0.99 \frac{\Omega_k - \langle \Omega_k \rangle}{\sigma_{\Omega_k}} \quad (R^2 = 0.97) \quad (14)$$

Table 9. Implied Ω_k from BAO c_0 and c_1

BAO coefficient	Implied Ω_k	$\sigma(\Omega_k)$
c_0	$+0.0051$	0.0009
c_1	$+0.0050$	0.0039

NOTE—Measurement errors from $\sigma(a_\alpha) = 1$ in whitened space.

After projecting out c_0 , the residual direction is almost perfectly correlated with curvature:

$$\frac{c_{1,\text{res}} - \langle c_{1,\text{res}} \rangle}{\sigma_{c_{1,\text{res}}}} = +0.99 \frac{\Omega_k - \langle \Omega_k \rangle}{\sigma_{\Omega_k}} \quad (r = 0.993) \quad (15)$$

Inverting the 1D regressions (Eqs. 14–15) using the BAO data projections a_0 and $a_{1,\text{res}}$ gives two independent estimates of Ω_k :

Both directions independently point to the same $\Omega_k \approx +0.005$ (Table 9)—a nontrivial internal consistency check. The BAO data’s departure from Λ CDM predictions is coherent across both measurable directions, as expected if the signal has a physical origin.

Both point to a mildly open universe ($\Omega_k > 0$), opposite to the CMB-preferred closed direction ($\Omega_k \approx -0.017$ in the ACT chain), but consistent with the positive Ω_k preferred by DESI BAO + Planck joint fits (Chen & Zaldarriaga 2025).

6.2. SN Blindness to Curvature

BAO SVD χ^2 vs Planck+DESI chain in (w_0, w_a) plane

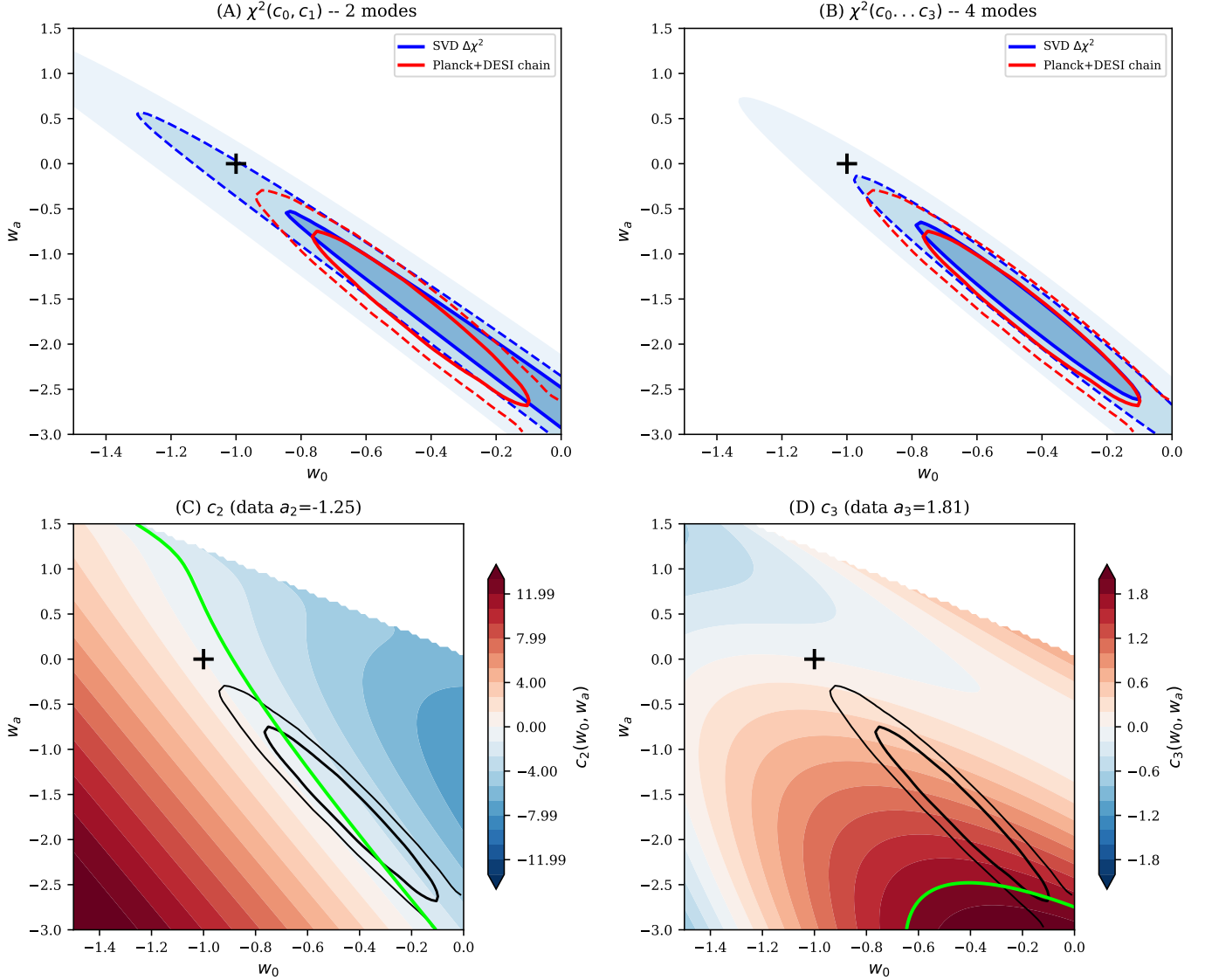


Figure 12. SVD χ^2 analysis in the (w_0, w_a) plane vs. the Planck+DESI chain posterior. (A) 2-mode SVD (blue) does not match chain (red). (B) 4-mode SVD closely matches. (C) c_2 colormap: the chain occupies a region of large $|c_2|$. (D) c_3 colormap: the chain's c_3 is consistent with the data. The point: four SVD modes reproduce the chain posterior, and the chain's displacement from Λ CDM is carried by c_0 and c_2 , not by the dark-energy mode c_1 .

SN probes cannot measure curvature: after projecting out c_0 , all three SN datasets give $\sigma_{\text{res}} < 0.1$, far below the measurability threshold. The BAO curvature coherence test therefore has no independent cross-check from SN data.

6.3. Effect on the c_0 Tension

Adding curvature makes the BAO c_0 tension *worse* (+2.6 σ vs. +2.2 σ in Λ CDM): the geometric degeneracy lets the CMB chain wander far from Λ CDM along the $\Omega_m h^2$ – Ω_k ridge, broadening the predicted c_0 distribu-

tion (Figure 14), but the broadening does not compensate for the larger mean offset between the chain and the BAO data. The Ω_k chain's c_0 distribution is much broader ($\sigma_{c_0} = 8.8$ vs. 1.6), reflecting the wide range of values allowed by the geometric degeneracy.

7. CMB-SIDE EXTENSIONS

We consider three CMB-side extensions that can shift the CMB posterior on $\Omega_m h^2$ without leaving distinct low-redshift distance signatures: (i) primordial magnetic fields (B_{PMF}), parametrized by the RMS field ampli-

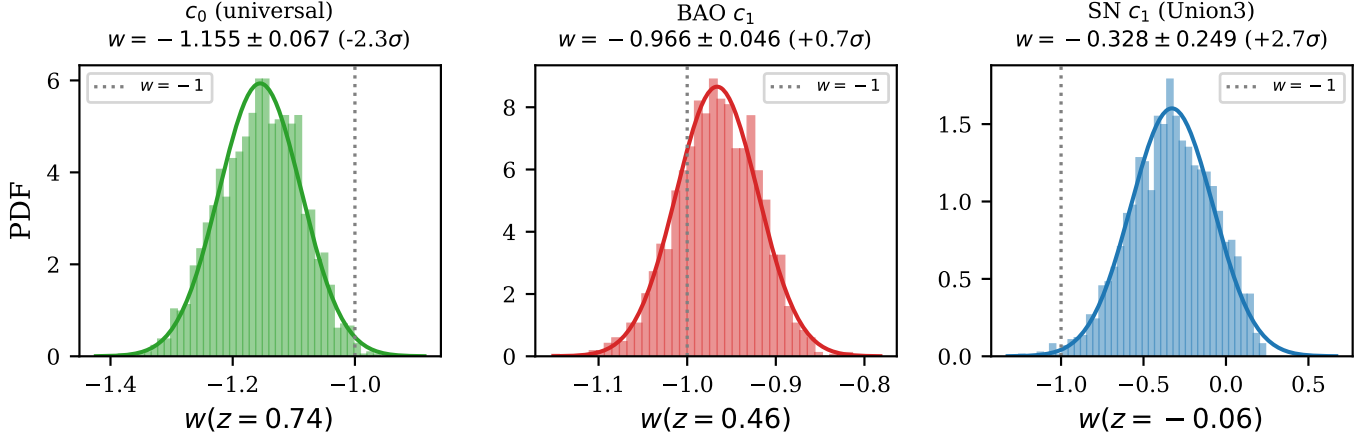


Figure 13. $w(z_{\text{pivot}})$ distributions. Colored histograms: Planck+DESI w_0 - w_a chain. Black dashed: data measurement from c_α projection (Gaussian with $\sigma_w = 1/|a_\alpha|$; for c_0 , σ includes $\Omega_m h^2$ marginalization). *Left:* c_0 pivot ($z = 0.74$)— $w \neq -1$, but this is the c_0 direction, which is affected by $\Omega_m h^2$. *Center:* BAO c_1 pivot ($z = 0.46$)—consistent with Λ CDM. *Right:* SN c_1 pivot ($z \approx 0$)—the chain predicts $w \approx -0.3$ but SN data gives $w \approx -1.3$, showing the chain’s DE extrapolation overshoots.

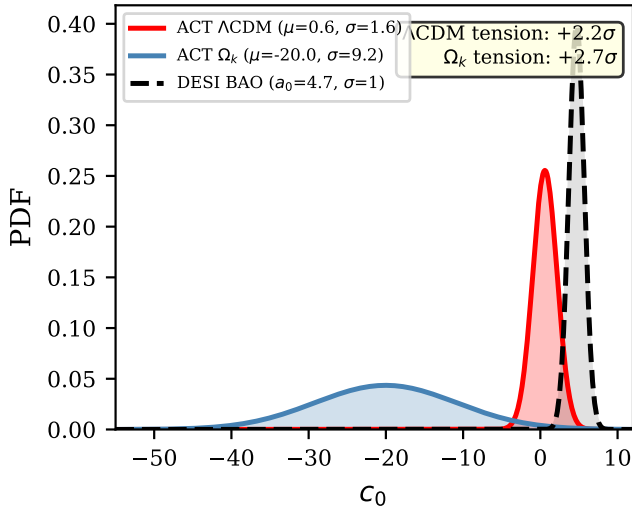


Figure 14. BAO c_0 distributions for Λ CDM (red) and Ω_k (blue) ACT chains, with DESI data (black dashed). The Ω_k distribution is much broader because the CMB geometric degeneracy allows $H_0 \in [55, 68] \text{ km s}^{-1} \text{ Mpc}^{-1}$. The broadening does not offset the larger offset of the chain mean from the data, so allowing curvature slightly *increases* the BAO c_0 tension ($2.2\sigma \rightarrow 2.6\sigma$).

tude on Mpc scales, which clump baryons and modifies how the universe recombines; (ii) early dark energy (EDE) with potential index $n = 2$ (the canonical Smith et al. (2020) flavor); and (iii) the phenomenological lensing rescaling A_{lens} , a multiplicative factor on the smoothing of CMB acoustic peaks by gravitational lensing ($A_{\text{lens}} = 1$ in Λ CDM (Calabrese et al. 2008)). Unlike spatial curvature (Section 6), these extensions do not open measurable new directions beyond c_0 .

7.1. How Extensions Shift c_0

Table 10. BAO c_0 Under CMB Extensions

Extension	$\langle c_0 \rangle$	σ_{c_0}	BAO tension	Notes
Λ CDM	+0.59	1.56	+2.2 σ	Baseline
A_{lens}	+2.36	1.86	+1.1 σ	Broadened
B_{PMF}	+2.04	2.37	+1.0 σ	Broadened
EDE $n = 2$	+0.85	1.88	+1.8 σ	$\approx \Lambda$ CDM

Each beyond- Λ CDM extension chain (ACT DR6 CMB-only) produces a different c_0 distribution:

Allowing A_{lens} or B_{PMF} to vary both broadens and shifts the CMB $\Omega_m h^2$ posterior, which reduces the BAO c_0 tension (Table 10; Figure 15). A_{lens} reveals that the CMB constraint on $\Omega_m h^2$ has a lensing component: in Λ CDM the amplitude of the acoustic peaks and the smoothing of those peaks by lensing are tied together (both depend on the primordial amplitude A_s and the late-time growth); freeing A_{lens} decouples the two and lets the $\Omega_m h^2$ marginal posterior shift.

A natural reading of this result is in terms of the optical depth uncertainties. Sailer et al. (2026) have shown that within Λ CDM, dropping Planck’s low- ℓ polarization and letting the remaining CMB + DESI DR2 BAO + CMB lensing data determine τ themselves yields $\tau = 0.090 \pm 0.012$, in ~ 3 – 5σ tension with the low- ℓ measurement $\tau = 0.054 \pm 0.007$. Thus this is one mechanism that would remove the tension that drives most of the preference of dynamical dark energy. One can make a back-of-envelope estimate to understand the mechanism. The CMB lensing reconstruction con-

Table 11. Residual Directions for CMB Extensions

Extension	Probe	σ_{res}	Meas.?	Tension
EDE $n = 2$	BAO	0.14	×	—
A_{lens}	BAO	0.07	×	—
B_{PMF}	BAO	0.16	×	—

strains $A_s (\Omega_m^{0.6} h)^{2.3}$ to $\sim 3\%$ (Planck 2015 XV, Eq. 14; Planck Collaboration et al. 2016), improved to $\sim 2\%$ with Planck PR4 / ACT DR6 (Eq. 49; Qu et al. 2024). The CMB temperature anisotropies determine $A_s e^{-2\tau}$ to $\sim 0.3\%$; combined with the low- ℓ polarization constraint $\sigma_\tau \simeq 0.006$, this gives $\sigma(\ln A_s) \simeq 1.2\%$. Using $\Omega_m^{0.6} h = \omega_m^{0.8} (\Omega_m h^3)^{-0.2}$ and the $\sim 0.3\%$ acoustic-scale determination of $\Omega_m h^3$, the implied lensing-only constraint on $\Omega_m h^2$ is $\sim 1.2\text{--}1.9\%$. The primary-anisotropy constraint on $\Omega_m h^2$ is $\sim 0.8\%$, so the lensing contribution is comparable to the primary such that when the central values disagree the joint posterior peak can shift by a non-negligible fraction of σ_{primary} . It motivates the need to improve the precision of the low- ℓ polarization measurements.

EDE $n = 2$ barely changes c_0 : its predicted distribution is indistinguishable from ΛCDM in c_0 (Table 10).

7.2. No New Measurable Directions

None of these extensions opens a measurable new direction (Table 11). The SVD residual cannot distinguish EDE, A_{lens} , and B_{PMF} from each other or from ΛCDM , because none of these extensions moves the low- z prediction measurably beyond V_0 .

8. DISCUSSION AND CONCLUSIONS

We have applied Singular Value Decomposition to DESI DR2 BAO measurements and three Type Ia supernova compilations (Union3, Pantheon+, DES-Dovekie), all analyzed with a common CMB baseline from ACT DR6. Our main findings are:

1. **Universal c_0 .** All low- z distance probes measure essentially the same leading SVD direction V_0 (cross-probe inner products > 0.996); we call its data amplitude c_0 . Within ΛCDM , this single number captures the leading low- z distance information from each probe — c_0 is, to high accuracy, a measurement of $\Omega_m h^2$.
2. **The $w_0\text{--}w_a$ preference is primarily c_0 .** The DESI $w_0\text{--}w_a$ preference is dominated by c_0 ; the

BAO dark energy direction (c_1) shows only $+1.2\sigma$ tension with ΛCDM . At the BAO c_1 pivot ($z = 0.46$), $w = -0.94 \pm 0.052$ ($+1.2\sigma$)—consistent with a cosmological constant. SN c_1 tensions are also mild (-1.5σ to $+0.2\sigma$). BAO prefers lower $\Omega_m h^2$ than the CMB ($+2.2\sigma$). The three SN datasets (DES-Dovekie -0.8σ , Pantheon+ -1.1σ , Union3 -1.7σ) do not have enough constraining power in c_0 to either confirm or refute the BAO result.

3. **Curvature coherence.** Spatial curvature is the only other extension that opens a new BAO direction, and only marginally. Both BAO measurable directions (c_0 and c_1) independently prefer positive spatial curvature — an interesting internal consistency check — although the c_1 error bar is very large. SN data give $\sigma_{\text{res}} < 0.1$ for Ω_k and so cannot provide an independent cross-check.
4. **No other extension is distinguishable.** No other tested extension (EDE, A_{lens} , B_{PMF}) opens a measurable new direction ($\sigma_{\text{res}} < 0.3$). Low- z distance data cannot distinguish among them.
5. **CMB lensing contribution and τ .** The CMB $\Omega_m h^2$ constraint has a lensing component: freeing A_{lens} both broadens and shifts the $\Omega_m h^2$ posterior, reducing the BAO c_0 tension from 2.2σ to 1.1σ . This is the same physics as shifting the optical depth τ , which Sailer et al. (2026) have shown the rest of the CMB + BAO data prefer to be $\sim 3\text{--}5\sigma$ higher than the Planck low- ℓ value ($\tau = 0.090 \pm 0.012$ versus 0.054 ± 0.007). At the current level of precision, a systematic in Planck’s τ would be a single-point failure for many cosmological analyses.
6. **The lowest-redshift BAO point.** A striking feature of the BAO data is the high D_M/D_H value of the lowest-redshift anisotropic point. Fitting it requires the higher SVD modes (c_2 , c_3) beyond the leading dark-energy direction (Figure 11, right panel), which within a dark-energy interpretation translates into a rapid evolution of $w(z)$. This makes that single point unusually influential. Any improvement in the next DESI data release—a smaller error bar on this point, or a measurement bringing comparable-precision information from lower redshift—would therefore be especially valuable for establishing whether this feature is real.

The SVD framework tells us *where* the tension is (c_0 , i.e., $\Omega_m h^2$) but not *what* causes it. This is a fundamen-

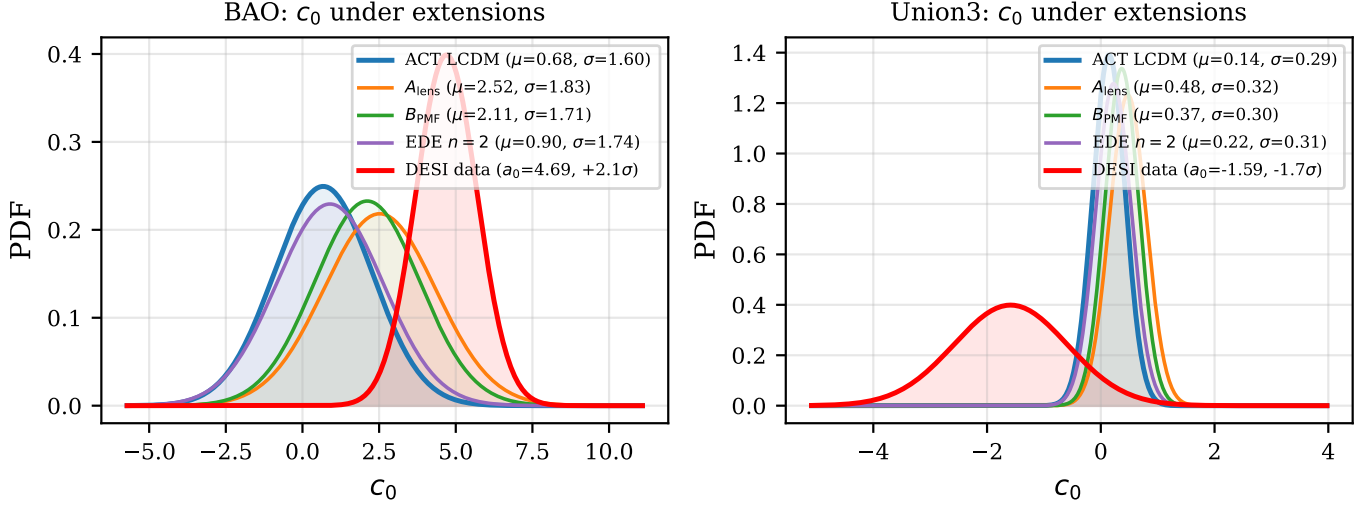


Figure 15. BAO c_0 distributions under beyond- Λ CDM extension chains. Blue: ACT Λ CDM. Colored: extension chains projected onto the same c_0 direction. Red dashed: DESI data ($\sigma = 1$). The right panel shows Union3 for comparison. The point: freeing A_{lens} or B_{PMF} broadens and shifts the predicted c_0 toward the data and so reduces the BAO tension, whereas EDE leaves it essentially unchanged.

tal limitation of distance measurements, not a shortcoming of the method: the SVD transparently identifies the boundary between what low- z probes can and cannot constrain.

All current low-redshift distance data, whether from BAO or supernovae, are effectively one-dimensional in

their cosmological information content. That single dimension is well described by $\Omega_m h^2$.

ACKNOWLEDGMENTS

This work was done entirely using Claude Code. MZ acknowledges support from the National Science Foundation NSF-BSF 2207583 and NSF 2209991, the Nelson Center for Collaborative Research and the Simons Foundation through the Black Holes and Strong Gravity program through Award No. SFI-MPS-BH-00012593-10.

APPENDIX

A. WHY c_0 IS DOMINATED BY $\Omega_m h^2$

This appendix shows that the β -vector and R^2 values reported in Section 3.1 can be reproduced analytically from the response of a single-redshift distance proxy to the CMB parameters, together with the 3×3 CMB correlation matrix. The result is presented for intuition: the SVD analysis in the main text uses the full 13-point BAO and 22-point SN datasets, not the simplified single-redshift proxy here.

A.1. Proxy Observables and Their Derivatives

We use two proxy observables to represent the full 13-point BAO and 22-point SN datasets. Throughout this appendix $D_M(z) \equiv c \int_0^z dz' / H(z')$ is the comoving distance defined in Section 2.1 (the line-of-sight and transverse distances coincide for the flat models used here):

- **BAO proxy:** $D_M(0.5)/r_d$, the comoving distance at $z = 0.5$ in units of the drag-epoch sound horizon r_d .
- **SN proxy:** $D_M(0.3)/D_M(0.5)$, the ratio of comoving distances at two redshifts. This ratio is independent of r_d and of the absolute luminosity M_B , capturing only the *shape* of the distance-redshift relation that SN constrain after M_B marginalization.

For each proxy, we compute logarithmic derivatives $g_p = \partial \ln(\text{obs}) / \partial \ln p$ for $p \in \{\Omega_m h^2, \Omega_b h^2, \theta_\star\}$ by finite differences around the fiducial cosmology. All derivatives are taken along the $\theta_\star$ -fixed slice—for each shift in $(\Omega_m h^2, \Omega_b h^2)$ we

Table 12. Log derivatives, fractional errors, and per-sigma sensitivities for the BAO proxy $D_M(0.5)/r_d$. The bottom two rows show the SN proxy log derivatives for comparison.

	$\Omega_m h^2$	$\Omega_b h^2$	θ_*
<i>BAO proxy: $D_M(0.5)/r_d$</i>			
$g_p = \partial \ln(D_M/r_d)/\partial \ln p$	+0.79	−0.26	−4.07
$\sigma_p/\langle p \rangle$	0.81%	0.51%	0.025%
$f_p (\times 10^3)$	+6.4	−1.3	−1.0
$ f_p/f_{\Omega_m h^2} $	1	0.21	0.16
<i>SN proxy: $D_M(0.3)/D_M(0.5)$</i>			
g_p	+0.10	−0.04	−0.41
$f_p (\times 10^3)$	+0.84	−0.20	−0.10

NOTE—The dominance of $\Omega_m h^2$ arises from the combination of a moderately larger log derivative ($|g_{\Omega_m h^2}|/|g_{\Omega_b h^2}| \approx 3$) and a moderately larger fractional error ($1.6\times$). The θ_* derivative is the largest in magnitude, but its tiny fractional error (0.025%) suppresses its contribution. The $D_M(0.3)/D_M(0.5)$ derivatives are smaller because H_0 cancels in the ratio.

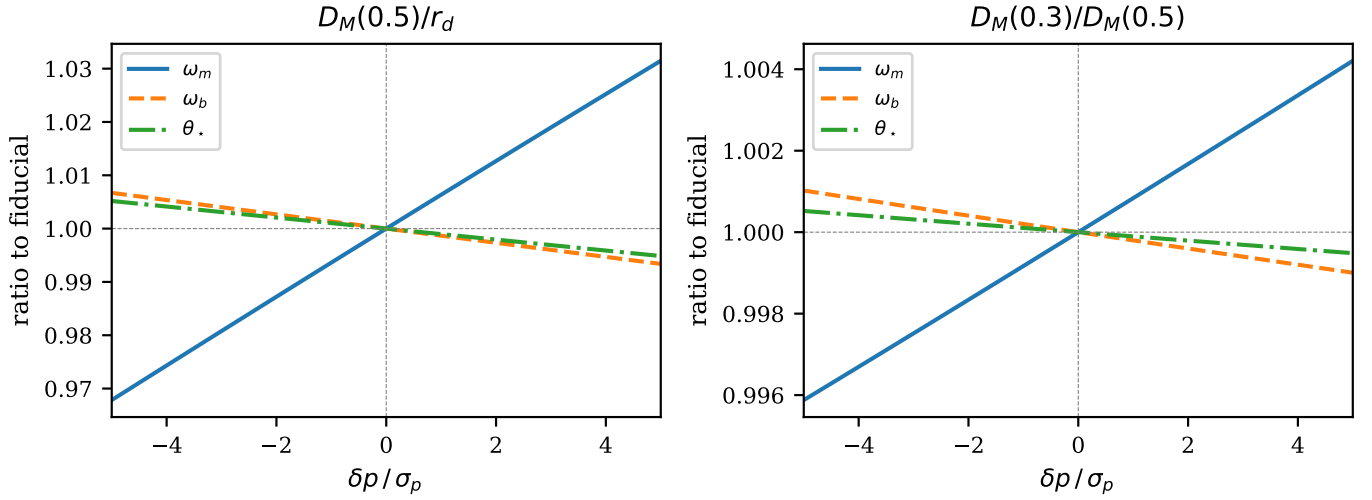


Figure 16. Proxy observables as a function of each CMB parameter (in units of σ_p from the ACT chain). Left: BAO proxy $D_M(0.5)/r_d$. Right: SN proxy $D_M(0.3)/D_M(0.5)$. All curves computed at fixed θ_* (solving for H_0), except the θ_* curve which varies the angular scale constraint. The slope at the origin times $\sigma_p/\langle p \rangle$ gives f_p (Table 12). The much steeper response to $\Omega_m h^2$ than to $\Omega_b h^2$ or θ_* is what makes c_0 predominantly an $\Omega_m h^2$ mode.

solve for the H_0 that keeps θ_* at its fiducial value—reflecting that the CMB pins the angular scale of the sound horizon orders of magnitude more tightly than H_0 itself. The θ_* scan instead varies θ_* explicitly.

The per-sigma sensitivity f_p combines the log derivative with the CMB fractional error:

$$f_p \equiv g_p \times \frac{\sigma_p}{\langle p \rangle}. \quad (\text{A1})$$

Table 12 lists g_p , σ_p^{frac} , and f_p for the BAO proxy. Figure 16 shows the observable responses graphically.

A.2. From Derivatives to the β -Vector

We *posit*—as a one-parameter ansatz, to be checked against the SVD output—that within the CMB posterior, c_0 depends linearly on the proxy observable. Writing the dependence in sigma-normalized parameters $\hat{p} = (p - \langle p \rangle)/\sigma_p$, this means

$$c_0 = -\alpha \sum_p f_p \hat{p} + \text{const}, \quad (\text{A2})$$

Table 13. Predicted vs. observed $\beta/\sigma(c_0)$ and R^2 .

Quantity	$D_M(0.5)/r_d$	$D_M(0.3)/D_M(0.5)$	Λ CDM (ACT chain)
$\beta_{\Omega_m h^2}/\sigma$	−0.88	−0.88	−0.90
$\beta_{\Omega_b h^2}/\sigma$	+0.18	+0.21	+0.13
β_{θ_*}/σ	+0.14	+0.11	+0.17
$R^2(\Omega_m h^2)$	0.94	0.94	0.95
$R^2(\Omega_m h^2 + \Omega_b h^2)$	0.98	0.99	0.97
$R^2(\text{all } 3)$	1.00	1.00	1.00

NOTE—Both proxies predict $\beta_{\Omega_m h^2}/\sigma$ to within 2%. The $\Omega_b h^2$ and θ_* coefficients differ by $\sim 30\%$ because the baryon loading of r_d and the θ_* sensitivity vary with redshift, and a single-redshift proxy cannot capture this. The key result $R^2(\Omega_m h^2) \approx 0.95$ is reproduced by both proxies.

for some proportionality constant $\alpha > 0$ that we will not need to compute. The main text reports the regression coefficients in the standardized form $\beta_p/\sigma(c_0)$, where $\beta_p = -\alpha f_p$ is the regression slope on $\hat{p}$ and $\sigma(c_0) = \alpha \sqrt{\mathbf{f}^\top \boldsymbol{\rho} \mathbf{f}}$ is the chain standard deviation of c_0 implied by the ansatz, with $\boldsymbol{\rho}$ the 3×3 correlation matrix of $(\Omega_m h^2, \Omega_b h^2, \theta_*)$. The ratio

$$\frac{\beta_p}{\sigma(c_0)} = \frac{-f_p}{\sqrt{\mathbf{f}^\top \boldsymbol{\rho} \mathbf{f}}}, \quad (\text{A3})$$

is manifestly independent of α : we cannot predict the absolute scale of c_0 from the proxy (it depends on the full SVD normalization), but we *can* predict which combination of CMB parameters dominates the chain variation of c_0 —and that is what the paper compares against. For the ACT chain:

$$\boldsymbol{\rho} = \begin{pmatrix} 1 & -0.30 & -0.21 \\ -0.30 & 1 & +0.26 \\ -0.21 & +0.26 & 1 \end{pmatrix}. \quad (\text{A4})$$

A.3. Predicted $\beta/\sigma(c_0)$ and R^2

The R^2 when regressing on the first k sigma-normalized parameters is also α -free:

$$R_k^2 = \frac{[\boldsymbol{\rho} \mathbf{f}]_k^\top \boldsymbol{\rho}_{kk}^{-1} [\boldsymbol{\rho} \mathbf{f}]_k}{\mathbf{f}^\top \boldsymbol{\rho} \mathbf{f}}, \quad (\text{A5})$$

where $[\boldsymbol{\rho} \mathbf{f}]_k$ denotes the first k components of $\boldsymbol{\rho} \mathbf{f}$ and $\boldsymbol{\rho}_{kk}$ is the $k \times k$ upper-left submatrix of $\boldsymbol{\rho}$. This is manifestly ≤ 1 (a squared projection) and equals unity when k equals the total number of parameters.

Table 13 compares the predictions from both proxies with the chain regression.

A.4. Physical Interpretation

The dominance of $\Omega_m h^2$ follows from three effects, all quantified above:

(i) *Larger per-sigma sensitivity.* For $D_M(0.5)/r_d$, $|f_{\Omega_m h^2}|$ is $4.8\times$ larger than $|f_{\Omega_b h^2}|$ and $6.2\times$ larger than $|f_{\theta_*}|$ (Table 12). This arises from the product of two moderate ratios: $|g_{\Omega_m h^2}|/|g_{\Omega_b h^2}| \approx 3$ (the log derivative) times $\sigma_{\Omega_m h^2}^{\text{frac}}/\sigma_{\Omega_b h^2}^{\text{frac}} \approx 1.6$ (the fractional error). Neither factor alone is dramatically large, but their product gives a factor ~ 5 .

(ii) *Chain correlations.* The negative correlation $\rho(\Omega_m h^2, \Omega_b h^2) = -0.30$ means that when we regress c_0 on $\Omega_m h^2$ alone, the regression coefficient absorbs part of the $\Omega_b h^2$ effect through the correlated component. Quantitatively, Eq. (A5) predicts $R^2(\Omega_m h^2) = 0.94$: the $\Omega_m h^2$ -only regression captures not just the direct $\Omega_m h^2$ sensitivity but also the portion of $\Omega_b h^2$ and θ_* variation that is correlated with $\Omega_m h^2$.

(iii) *SN and M_B marginalization.* For SN, which marginalize over the absolute magnitude M_B , the relevant observable is the *shape* of the distance–redshift relation, captured here by the ratio $D_M(0.3)/D_M(0.5)$. Its log derivatives are much smaller than the BAO proxy’s ($g_{\Omega_m h^2} = +0.10$ versus $+0.79$ for D_M/r_d), so the SN proxy responds far less in absolute terms. But the quantity the paper compares, $\beta_p/\sigma(c_0)$, is a *unit-normalized* vector (Eq. A3): dividing by $\sigma(c_0)$ scales out the overall size of the response and leaves only *which* parameter is most responsible for the chain

scatter of c_0 , not how large that scatter is. Because the normalization $\sqrt{\mathbf{f}^\top \boldsymbol{\rho} \mathbf{f}}$ shrinks in proportion to the smaller $\mathbf{f}$, the *direction* of the β -vector comes out essentially the same as for BAO — which is why both probes identify c_0 with the same $\Omega_m h^2$ combination, even though SN constrain its amplitude far more weakly.

The SN sensitivity to $\Omega_m h^2$ survives M_B marginalization for a physical reason worth stating. All derivatives are taken at fixed θ_* , which links H_0 to $(\Omega_m h^2, \Omega_b h^2)$ through $\theta_* = r_*/D_M(z_*)$. Increasing $\Omega_m h^2$ then *decreases* H_0 (to hold the distance to last scattering fixed), which amplifies the induced change in $\Omega_m = \Omega_m h^2/h^2$. Evaluated at fixed θ_* , $\partial \ln \Omega_m / \partial \ln \Omega_m h^2|_{\theta_*} \approx 2.6$ (computed in the reproducible pipeline and confirmed by a direct regression of $\ln \Omega_m$ on $\ln \Omega_m h^2$ across the ACT Λ CDM posterior) — substantially larger than the naive value of unity that would hold at fixed H_0 . This is why even the M_B -marginalized SN distance *shape*, which depends on Ω_m rather than $\Omega_m h^2$ directly, retains substantial sensitivity to $\Omega_m h^2$ within the CMB posterior.

REFERENCES

- ACT Collaboration. 2025.
<https://arxiv.org/abs/2503.14454>
- ACT Collaboration, Louis, T., et al. 2025.
<https://arxiv.org/abs/2503.14452>
- Calabrese, E., Slosar, A., Melchiorri, A., Smoot, G. F., & Zahn, O. 2008, PhRvD, 77, 123531,
doi: 10.1103/PhysRevD.77.123531
- Chen, S.-F., & Zaldarriaga, M. 2025.
<https://arxiv.org/abs/2505.00659>
- Cortês, M., & Liddle, A. R. 2024, JCAP, 2024, 005.
<https://arxiv.org/abs/2404.08056>
- DESI Collaboration. 2025a.
<https://arxiv.org/abs/2503.14738>
- . 2025b. <https://arxiv.org/abs/2503.14739>
- Efstathiou, G. 2025a. <https://arxiv.org/abs/2505.02658>
- . 2025b. <https://arxiv.org/abs/2408.07175>
- Eisenstein, D. J., & White, M. J. 2004, PhRvD, 70, 103523,
doi: 10.1103/PhysRevD.70.103523
- Knox, L. 2006, PhRvD, 73, 023503,
doi: 10.1103/PhysRevD.73.023503
- Planck Collaboration. 2020, A&A, 641, A6.
<https://arxiv.org/abs/1807.06209>
- Planck Collaboration, Ade, P. A. R., et al. 2016, A&A, 594, A15. <https://arxiv.org/abs/1502.01591>
- Popovic, B., et al. 2025, arXiv e-prints.
<https://arxiv.org/abs/2511.07517>
- Qu, F. J., Sherwin, B. D., Madhavacheril, M. S., et al. 2024, ApJ, 962, 112. <https://arxiv.org/abs/2304.05202>
- Rubin, D., et al. 2023. <https://arxiv.org/abs/2311.12098>
- Sailer, N., Farren, G. S., Ferraro, S., & White, M. 2026, PhRvL, 136, 081002,
doi: 10.1103/PhysRevLett.136.081002
- Scolnic, D., et al. 2022, ApJ, 938, 113.
<https://arxiv.org/abs/2202.04077>
- Weiner, Z. J. 2026. <https://arxiv.org/abs/2603.18131>
- Ye, G., & Lin, S.-J. 2025. <https://arxiv.org/abs/2505.02207>